Distributed-delay formulations of a thermal laboratory model: identifiability, uncertainty, and experiment design

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Abstract

The Temperature Control Laboratory (TC-Lab) provides transient heater and sensor data for studying parameter estimation and experiment design. We compared n-stage input-delay models in conductance–capacity and gain–time-constant coordinates, separating formulation, initialization and numerical propagation. The original four-parameter thermal model has an exact one-dimensional physical parameter equivalence family under the modeled output and prescribed initial conditions. The cold one-stage formulation represents its transfer behavior with three parameters and a global pole-label ambiguity. Constant input with equilibrated delay states leaves delay unobservable. Coordinate changes alter conditioning without adding information. Cold joint calibration favors one stage, while initialization changes the order ranking and transfer errors on held-out records exceed calibration errors. Profiles, resampling and controlled coverage comparisons show strong sensitivity to serial-error assumptions; HAC intervals improve but do not restore nominal coverage in the longer correlation scenario. Ambient/history and command-hold sensitivity further limit the interpretation of narrow conditional intervals. Constrained local input designs improve their declared information and fixed-mean separation criteria. Refitting competitors removes equilibrated constant-input separation and qualifies fixed-mean discrimination claims. The results support identifiable transfer functions and explicit experimental assumptions as the basis for inference. This draft and all scientific choices await human review.

1 Introduction and theoretical grounding

Parameter estimation for the TC-Lab must distinguish uncertainty in a measured thermal response from ambiguity in its physical explanation. A model may predict the observed temperature closely while several parameter combinations produce the same response. Changing parameter coordinates can improve numerical scaling, but an invertible coordinate transformation preserves that ambiguity.

Continuous-time identification also distinguishes the input-driven model from the residual process. The latter can include measurement error, unmeasured disturbances, and model inadequacy (Garnier et al., 2008). A fitted residual covariance therefore does not by itself identify sensor noise or establish a physical disturbance mechanism. This distinction guides the uncertainty comparisons below: we vary working noise assumptions and examine the errors that remain on experiments excluded from calibration.

Identifiability concerns the uniqueness of parameters given a model, an observed output, and an experiment. Practical estimability additionally depends on the record length, input, noise, and initial-state information. Profile likelihood examines a parameter while reoptimizing the remaining parameters and can reveal features missed by a local quadratic approximation (Raue et al., 2009). We

combine that approach with explicit transfer functions, sensitivity singular values, and resampling. The algebraic calculation is needed to distinguish an exact equivalence class from a numerically shallow objective or a finite exploration bound.

Linear time-invariant system identification supplies a connection between input excitation, parameter information, and prediction accuracy. Application-oriented design can target informative parameter combinations rather than require equal precision in every physical parameter. Hjalmarsson (2009) develops this distinction using parameter confidence regions and application loss. Its prediction-error information formula assumes that the true system belongs to the model set and that the predictor is differentiable. Those assumptions require separate scrutiny here because the TC-Lab residuals can contain serial structure and between-experiment discrepancies.

Finite-horizon amplitude constraints introduce a further distinction between an information objective and its numerical solution. Manchester (2010) formulates time-domain information design as a nonconvex input problem and develops a convex relaxation. We use bounded multistart nonlinear optimization with explicit command and temperature constraints. We have not implemented that relaxation, so its approximation guarantees do not apply to the designs reported here. Comparing local solutions across assumed models and residual covariances is part of assessing a proposed experiment.

An n-stage chain is a family of distributed-delay models. The linear chain construction relates cascaded first-order stages to Erlang dwell times (Hurtado and Kirosingh, 2019). For a fixed mean delay, increasing the stage count narrows the delay distribution and approaches a pure delay. Fitting this family therefore tests a shape assumption as well as a numerical approximation. Its auxiliary stage count differs from its number of estimated parameters.

This study asks how these distinctions affect the available TC-Lab data. We compare physical and input–output parameterizations, quantify conditional uncertainty, test sensitivity to initialization and data handling, and evaluate simulated inputs for future estimation and discrimination. The study is a reproducible draft for internal review; it does not establish a validated physical mechanism from temperature fit alone.

Appendix D derives the frequency sensitivities of the delayed models and an ideal periodic information matrix. It connects the low-frequency lag ambiguity and repeated-pole singularity to the information criteria used in the finite-horizon designs.

2 Model formulations

2.1 Thermal balance with a distributed input delay

The recorded input $u(t)$ is the heater command in percentage points, and the measured output $T(t)$ is the first temperature sensor in degrees Celsius. We fixed the heater conversion $c = \alpha P$ at 0.032 W per percentage point, using the inherited values $\alpha = 0.00016$ and $P = 200$. The ambient temperature T_{amb} was set to the first measured temperature in each experiment. This choice conditions the analysis on an uncertain ambient estimate.

For a chain of n delay states, the auxiliary variables $z_i(t)$ have the same units as $u(t)$. The physical parameters $U_{\mathrm{a}} > 0$, $C_{\mathrm{p}} > 0$, and $\theta > 0$ denote thermal conductance, effective heat capacity, and mean input-delay time, respectively. The model is

$$\dot{z}_1 = \frac{n}{\theta}(u - z_1), \qquad \dot{z}_i = \frac{n}{\theta}(z_{i-1} - z_i), \quad i = 2, \ldots, n, \tag{1}$$

$$C_{\mathrm{p}}\dot{T} = -U_{\mathrm{a}}(T - T_{\mathrm{amb}}) + cz_n. \tag{2}$$

The input–output (IO) coordinates $K = c/U_a$ and $\tau = C_p/U_a$ are steady-state gain and thermal time constant. In these coordinates the temperature equation is

$$\dot{T} = -\frac{T - T_{amb}}{\tau} + \frac{K}{\tau} z_n. \tag{3}$$

Both parameterizations describe the same positive model family. Their inverse map is $U_a = c/K$ and $C_p = c\tau/K$. Distinct rectangular parameter bounds need not map onto each other; our numerical comparison records active bounds and compares common interior solutions.

For zero initial deviations, the Laplace variable s gives the transfer function

$$G_n(s) = \frac{K}{(1 + \tau s)(1 + \theta s/n)^n}. \tag{4}$$

The delay chain has an Erlang impulse-response kernel with mean θ and variance θ^2/n; Appendix A derives these quantities. The linear chain construction is consistent with the Erlang dwell-time representation studied by Hurtado and Kirosingh (2019). A finite chain responds before time θ and therefore represents a distributed lag. As n increases at fixed θ, the delay factor $(1 + \theta s/n)^{-n}$ tends to $\exp(-\theta s)$. We also fitted the limiting pure-delay model,

$$\tau \dot{T}(t) = -(T(t) - T_{amb}) + K u(t - \theta). \tag{5}$$

Increasing n changes the transfer-function shape and adds dynamic states; it does not add estimated parameters to the three-parameter family.

2.2 Initial-state assumptions

We conditioned $T(0)$ on the first observation. The cold-start assumption set all $z_i(0) = 0$, while the input-equilibrated assumption set all $z_i(0) = u(0)$. The pure-delay model used the corresponding zero or constant input prehistory. These assumptions encode different unobserved histories of heater operation. Neither was established by the recorded input before the first sample, because that history is absent from the supplied data. Input-equilibrated refers to delay memory. Under a constant command u_0, equilibrium of the entire model would additionally require $T(0) = T_{amb} + K u_0$. Here the thermal initial condition is fixed independently at the first reading. Determining the preceding thermal trajectory requires pre-window observations.

2.3 Original two-state thermal model

The inherited model describes heater temperature T_H and sensor temperature T_S. Its conductances $U_a, U_b > 0$ and heat capacities $C_{p,H}, C_{p,S} > 0$ satisfy

$$C_{p,H}\dot{T}_H = -U_a(T_H - T_{amb}) - U_b(T_H - T_S) + cu, \tag{6}$$

$$C_{p,S}\dot{T}_S = U_b(T_H - T_S). \tag{7}$$

Only T_S was used in this single-output comparison. Both initial temperatures were fixed at the first sensor reading. Under these conditions its transfer function depends on three combinations of four physical parameters:

$$G_{orig}(s) = \frac{K}{1 + as + bs^2}, \qquad K = \frac{c}{U_a}, \quad a = \frac{C_{p,H}}{U_a} + \frac{C_{p,S}}{U_b} + \frac{C_{p,S}}{U_a}, \quad b = \frac{C_{p,H}C_{p,S}}{U_a U_b}. \tag{8}$$

Appendix A exhibits the resulting continuum of equivalent physical parameters. The one-stage cold-start model has $a = \tau + \theta$ and $b = \tau\theta$, so its zero-state transfer family coincides with this original model when the poles are distinct. The one-stage model also permits a repeated pole, which is a limiting case of the strictly positive original family: the original family has $a^2 > 4b$, whereas the repeated pole has $a^2 = 4b$. The pole labels τ and θ can be interchanged. We used the convention $\tau \geq \theta$ when the exchanged point remained within the specified parameter bounds and retained both modes in likelihood profiles.

3 Computational methods

3.1 Data and reproducibility

We preserved the supplied observations and first reproduced the unchanged inherited simulations twice. The resulting trajectories were byte-identical within each dataset. We then exported the teaching notebooks to Python scripts with cell markers and source hashes, checking equality of their scientific abstract syntax trees. The delay study uses separate Python modules for data, simulation, inference, and result generation.

The data audit found 15 CSV files. Two validation files were numeric duplicates of the sine and step calibration files, respectively. We excluded both duplicates from validation, leaving 11 files. We fitted the sine experiment, the step experiment, and their combination separately. Each calibration experiment has 901 temperature observations. The first observation conditions the initial temperature and is excluded from the fitted residual vector, leaving 900 residuals per experiment. This treatment does not propagate uncertainty in the initial temperature or in the ambient temperature inferred from it.

Long calculations run from immutable copies of committed source and raw data. The run manifests retain source, configuration and data hashes, package versions, random seeds, every optimizer attempt, and failures. A single manifest identifies each numerical phase. Result tables and figures are generated from saved outputs.

3.2 Simulation and fitting

We used exact state propagation of the linear time-invariant dynamics over each sampling interval, with linear interpolation of the recorded heater command. Separate checks compared left and right input holds and the inherited backward Euler discretization. The pure-delay simulator integrates across shifted input knots. These choices distinguish input reconstruction and numerical discretization from a change in model equations.

The baseline comparison fitted seven orders, $n \in \{1, 2, 3, 4, 6, 8, 12\}$, two initial-state assumptions, and both parameter coordinates. Pure-delay fits used the same two initial-state assumptions and coordinates. The original model used its four physical parameters. These choices gave 99 calibration cases. We minimized unweighted temperature sum of squared errors (SSE) using bounded least squares in logarithmic parameters, with 24 starts per case and seed 532. The broad input–output bounds were $0.2 \leq K \leq 1.5$, $5 \leq \tau \leq 600\,\mathrm{s}$, and $0.01 \leq \theta \leq 200\,\mathrm{s}$. The physical-coordinate bounds were $0.032/1.5 \leq U_\mathrm{a} \leq 0.032/0.2\,\mathrm{W\,K^{-1}}$, $0.1 \leq C_\mathrm{p} \leq 100\,\mathrm{J\,K^{-1}}$, and the same delay bounds. The original model used $U_\mathrm{a} \in [0.005, 0.3]$ and $U_\mathrm{b} \in [0.0001, 1]$ in $\mathrm{W\,K^{-1}}$, with $C_\mathrm{p,H} \in [0.01, 100]$ and $C_\mathrm{p,S} \in [0.001, 20]$ in $\mathrm{J\,K^{-1}}$. The reduced physical rectangle is not the exact image of the IO rectangle. Same-point covariance and information comparisons separate coordinate effects from changes in the feasible parameter domain. We retained convergence status and active bounds rather than

interpreting every optimizer return as a successful estimate. Root mean squared error (RMSE) uses the same retained residuals.

3.3 Ambient, history and input-reconstruction sensitivity

We prespecified 241 additional calibrations. For each of eight delayed models and three calibration datasets, four nuisance treatments give 96 fits: free ambient with cold history, free ambient with equilibrated history, free history with fixed ambient, and both free. Ambient offsets have bounds $[-2, 2]\,^{\circ}\mathrm{C}$, separately for each record; history fractions γ_j have bounds $[0, 1]$. These bounds describe assumed sensitivity ranges. They are not independently measured probability distributions. The three IO parameters are shared across records in a joint fit. We use logarithmic IO coordinates, additive ambient offsets and untransformed history fractions, with analytic mean derivatives, 24 starts, seed 532 and at most 1,200 function evaluations per start. Appendix B derives exact ambiguities that can remain when these quantities are free.

Another 96 joint fits hold ambient offsets at $-1, -0.5, -0.25, 0.25, 0.5, 1\,^{\circ}\mathrm{C}$ for each delayed model and both prescribed histories. For these fixed perturbations, the same offset and history assumption are applied to each held-out record. Fitted record-specific nuisance quantities are not assigned automatically to an unseen record; those fits have no corresponding validation correction.

The remaining 49 joint fits compare input reconstruction and discretization: 32 exact left/right-hold fits across eight delayed models and two startups, 14 right-hold backward-Euler fits across the seven finite chains and two startups, and three original-model variants (exact left hold, exact right hold, and right-hold backward Euler). They use the same 24-start budget of 1,200 function evaluations. We retain active bounds, numerical ranks, all attempts and any failure to recover the nested fixed-assumption objective. Reduced calibration SSE after adding nuisance parameters supplies no independent validation evidence.

3.4 Local uncertainty and estimability

For residual vector $\boldsymbol{e}$ and log-parameter vector $\boldsymbol{\eta}$, the sensitivity matrix $J = \partial \boldsymbol{e}/\partial \boldsymbol{\eta}$ defines the conditional information matrix $F = J^{\top} J/\sigma^2$ under independent and identically distributed (IID) Gaussian errors of variance σ^2. We computed the singular value decomposition of J directly and reported the eigenvalues and directions of F. Numerical rank uses a relative singular-value cutoff of 10^{-8}. This threshold is a numerical convention; the explicit equivalence classes in Appendix A establish the structural conclusions independently.

We reported local uncertainty under a fixed hypothetical sensor standard deviation of $0.25\,^{\circ}\mathrm{C}$ and under a common variance estimated from each calibration residual vector. The fixed value is a sensitivity scenario, not a calibration established by these experiments. Rank-deficient full parameter covariances are undefined. A diagnostic pseudoinverse is retained separately and is not used as a confidence interval for a null direction. We also computed heteroskedasticity and autocorrelation consistent (HAC) sandwich estimates at lag bandwidths of 10, 30, and 60 samples, together with residual autocorrelation functions (ACFs) and Ljung–Box statistics. The sandwich uses Bartlett lag weights (Newey and West, 1987). Its asymptotic justification requires zero-mean score moments, bounded moments, and suitable decay of dependence. Fixed bandwidths in these finite records do not establish those assumptions or nominal coverage. These calculations assess sensitivity to serial dependence; they do not correct a biased mean model.

3.5 A working correlated-residual likelihood

We also fitted the 48 IO calibration cases with stationary Gaussian Ornstein–Uhlenbeck (OU) residual covariance $R_{ij} = \sigma^2 \exp(-|t_i - t_j|/\ell)$ within each experiment, independently across experiments. Actual retained observation times are used. For residuals e_i, set $w_1 = e_1$ and

$$\phi_i = e^{-(t_i - t_{i-1})/\ell}, \qquad w_i = \frac{e_i - \phi_i e_{i-1}}{\sqrt{1 - \phi_i^2}}.$$

With $q = \sum_j \|\boldsymbol{w}_j\|^2$ over records and N retained observations, profiling the common variance gives $\hat{\sigma}^2 = q/N$ and negative log-likelihood

$$\mathcal{L} = \frac{1}{2}\left[N\{\log(2\pi) + 1 + \log(q/N)\} + \sum_j \sum_{i=2}^{N_j} \log(1 - \phi_{ij}^2)\right].$$

The determinant term is retained while optimizing the three log mean parameters and $\log \ell$, with $\ell \in [0.05, 3000]$ s, 24 starts and the same seed. All solver attempts and boundary diagnostics are retained. We compare conditional mean information at fixed fitted noise with observed curvature that profiles variance and includes mean–length coupling. Two Hessian steps, 10^{-4} and 5×10^{-5} in log coordinates, check numerical sensitivity. A fitted OU length or improved likelihood does not establish a residual mechanism; raw and filtered correlation diagnostics assess this working model.

3.6 Likelihood profiles and resampling

For each native parameter we fixed its value and reoptimized all nuisance parameters, following the profile construction of Raue et al. (2009). The exploration combined a global grid with local refinement, targeting 61 points and four nuisance starts at each point. The exact count after merging grids is recorded per profile. Additional points examined the exchanged pole mode of the one-stage cold-start model. We retained disconnected accepted components and endpoints at the exploration bounds. An endpoint at such a bound does not establish an unbounded mathematical interval.

For N residuals and sum of squared errors $S(\boldsymbol{\eta})$, the likelihood-ratio statistics were $[S(\boldsymbol{\eta}) - S(\hat{\boldsymbol{\eta}})]/\sigma^2$ for fixed variance and $N \log[S(\boldsymbol{\eta})/S(\hat{\boldsymbol{\eta}})]$ for a common profiled variance. The nominal 95% reference threshold was the 0.95 quantile of a chi-squared distribution with one degree of freedom. These thresholds assume an appropriate independent Gaussian likelihood and regular identification; they are conditional diagnostics when those assumptions fail.

After comparing first-pass sets with the exact physical equivalence family, we refined a declared selection of observed crossings in a separate phase. The selection covers the redundant original physical coordinates, cold one-stage delay profiles in both coordinates, and the representative joint cold higher-order delay profiles. Feasible exact-equivalence witnesses are included before crossing refinement. Bisection stops at a likelihood-ratio residual of 10^{-3} or a log-parameter bracket width of 10^{-7}, with at most 26 steps and four nuisance starts per new point. The original grids are preserved. A separate audit checks forward objectives at new points and whole compatible-family inclusion inside the fitting box; it cannot establish the absence of other modes.

The resampling design specified 400 refits for each of four circular residual block lengths (1, 10, 30, and 60 samples), plus 400 independent Gaussian simulations at the fixed 0.25 °C standard deviation, for each input–output calibration case. Residuals were centered and resampled within each experiment; first temperatures, ambient estimates, and heater calibration were fixed. Each draw was transformed exactly into physical coordinates. Structural null directions were marked undefined rather than assigned an interval from arbitrary optimizer selections. These resampling distributions

remain conditional on the mean model and on the assumed within-experiment stationarity. The fixed-length circular scheme retains equal marginal sampling weights at record edges. It differs from the geometric block lengths of the stationary bootstrap; Politis and Romano (1994, physical PDF p. 3, Remark 2) describes the fixed-length circular relative. Neither that finite-sample mean property nor the paper's stationary-bootstrap theory establishes coverage for our nonlinear fits with model discrepancy. We assess selected interval procedures separately using independent synthetic datasets.

The independent coverage design prespecified 300 synthetic datasets for each of four joint cold generating models (one, two and twelve stages, plus pure delay) and three true error processes: IID Gaussian, stationary OU with 10 s length, and stationary OU with 60 s length. Every process has marginal standard deviation 0.25 °C. We used the recorded input and observation times, fixed the conditioned initial temperatures exactly, and refitted each synthetic dataset before constructing local and profile intervals. The comparison includes known-variance and estimated-variance IID log-Wald intervals, three HAC bandwidths, and both delay-profile variance conventions. Physical intervals use the exact parameter transformation.

All 3,600 requested datasets remain in the coverage denominator. We report coverage among defined intervals, its Wilson binomial interval, and worst/best unconditional bounds that count failed or undefined outcomes as uncovered or covered, respectively. Failed worker chunks and missing result files are distinguished: a completed chunk with a missing file fails the integrity audit. Disconnected profile widths sum the accepted components; envelope widths also include excluded gaps. A secondary three-point quadratic interpolation checks whether finite-grid interpolation changes coverage classification at the known true delay. It does not replace nuisance reoptimization or certify endpoint accuracy. The study conditions on these generating points and does not estimate coverage under real between-record model discrepancy. Bootstrap intervals remain real-data sensitivity comparisons; their repeated-sampling coverage is unmeasured by this experiment.

3.7 Conditional prediction uncertainty

For target mean sensitivity $\boldsymbol{j}(t)$, local mean variance is $\boldsymbol{j}(t)C\boldsymbol{j}(t)^\top$. We use covariance on the identifiable quotient with residual degrees of freedom $N-\mathrm{rank}(J)$, checking that the target annihilates each structural null direction. Known global ambiguities require an additional check: an equilibrated constant-input calibration leaves delay free, so we withhold bands for a varying-input target even if its local delay derivative happens to vanish at the selected parameter. The original model's zero-state physical equivalence preserves predictions globally and permits finite prediction uncertainty on its quotient. The eleven rank-deficient baseline cases were the three original-model gauge cases and eight equilibrated constant-step cases. The synthetic repeated-pole example is excluded from this regular-quotient covariance interpretation.

Local pointwise normal mean bands are compared with percentile mean bands from saved joint-fit bootstrap trajectories. An additional independent future error law is required for observation prediction bands. We add its plug-in marginal variance under IID or working OU assumptions; HAC supplies mean bands without inventing a future error distribution. For Gaussian parametric bootstrap draws, future-observation quantiles are computed from the declared equal-weight mixture of 400 Gaussian trajectory components. This is a numerical mixture calculation, with the inferential accuracy of the bootstrap still conditional and unproven. Circular-block draws supply mean bands only. The first observation is excluded from all bands because it conditions the state. Existing-record pointwise containment is descriptive, without a repeated-experiment or simultaneous coverage interpretation.

3.8 Prospective experiment design

The prespecified input-design horizon is 900 s with 1 s observations and 19 command knots at 50 s intervals, joined linearly. Command amplitudes lie between 0 and 100%, their time average is 50%, and the slew limit is one percentage point per second. Ambient and initial measured temperature are fixed at 25 °C. We constrain the predicted mean temperature below 85 °C for all sixteen joint fitted mean/startup combinations, regardless of the model used for the utility. These are assumed simulation constraints, without a hardware safety certification or uncertainty margin. An independent 0.25 s forward simulation also checks a 300 s shutdown tail, ramping the last command to zero at the same slew limit. The first design trial met its 1 s constraints but exceeded the assumed limit slightly between observations in that denser check. We retained that trial and tightened the optimization bound by 0.01 °C for both parameter and discrimination designs. This numerical guard leaves the reported acceptance limit at 85 °C; every selected design must still pass the independent dense shutdown-tail check. The guard itself supplies no continuous-time bound or hardware guarantee.

For a future sensitivity matrix $J(u)$ and assumed covariance R, the local information is $F(u) = J(u)^\top R^{-1} J(u)$, optionally augmented by information from the two historical calibration experiments under the same model and covariance. This added information is not a Bayesian distribution over parameters. The four utilities are $\log\det F$, $\log\lambda_{\min}(F)$, $-\log[(F^{-1})_{\theta\theta}]$, and $-\log\operatorname{tr}(AF^{-1})$. Here A is the scenario-specific average thermal-mean sensitivity outer product for a future constant 50% reference. The variance target concerns mean prediction, without future measurement noise. The third utility targets log-delay variance. Null-space compatibility is checked before a target-specific pseudoinverse is used. Parameter metrics are natural logarithms of (K, τ, θ).

Nominal designs use cold one-stage, two-stage, twelve-stage and pure-delay fits with IID, OU10 s or OU60 s working correlations, each with marginal standard deviation 0.25 °C. Ensemble designs use all 48 model/startup/covariance scenarios. We aggregate scenario-relative log utilities by their mean or minimum. Four utilities and two historical-information settings give 112 planned designs. Six common command starts include constant, sine, multisine, random binary and alternating knot patterns, each satisfying the amplitude, dose and slew rules. Constant input supplies the normalization reference where its utility is finite; otherwise we use the multisine. Multistart sequential least squares programming (SLSQP) uses analytic input gradients and retains convergence and constraint violations; its local solutions have no global-optimality certificate. Minimum aggregation is nonsmooth at scenario ties.

The physical and IO log-coordinate information matrices are related by the congruence in Appendix A. Because $|\det M| = 1$, the log-determinant utility is equal at corresponding points. Delay variance and properly transformed prediction loss are also invariant. The smallest eigenvalue depends on the chosen metric. Thus a coordinate change can alter an E-criterion without changing the underlying prediction experiment.

We separate parameter precision from discrimination of stage count. For two fixed future means with the same assumed covariance, define

$$D_{ij}^2(u) = [\mu_i(u) - \mu_j(u)]^\top R^{-1} [\mu_i(u) - \mu_j(u)].$$

The Gaussian log-likelihood ratio has mean $\pm D_{ij}^2/2$ and variance D_{ij}^2 under these two simple hypotheses. Equal-prior minimum classification error is therefore $\Phi(-D_{ij}/2)$. This ideal calculation assumes fixed known means and covariance; it is not a testing-power estimate after parameter estimation or under TC-Lab discrepancy.

Twelve additional local designs aggregate relative log squared separation for all six pairs among one-stage, two-stage, twelve-stage and pure-delay means, at both startups and all three working

Table 1: Calibration RMSE in degrees Celsius, excluding the initial temperature. Each entry comes from its own fitted calibration/startup combination. Equilibrated means the prescribed delay states equal the initial command, with the sensor temperature fixed by the first observation.

Model	Sine, cold	Step, cold	Joint, cold	Joint, equilibrated
$n = 1$	0.2567	0.1573	0.4115	0.7936
$n = 2$	0.3470	0.1529	0.4630	0.7585
$n = 3$	0.3810	0.1573	0.4841	0.7486
$n = 4$	0.3985	0.1610	0.4953	0.7441
$n = 6$	0.4165	0.1659	0.5070	0.7398
$n = 8$	0.4256	0.1689	0.5130	0.7378
$n = 12$	0.4348	0.1724	0.5192	0.7359
Pure delay	0.4541	0.1812	0.5323	0.7322

covariances. The multisine supplies the normalizing reference. A bounded 72-case diagnostic then lets each alternative model refit its three parameters against a fixed one-stage generating mean, using 24 starts under the same future covariance. It compares constant, multisine, and both selected discrimination designs against two-stage, twelve-stage and pure-delay alternatives. This directional subset does not cover every generating model or parameter point. The one-stage order was chosen as a representative because it has the smallest cold joint calibration error; this choice does not establish it as the true system. A converged refit distance is an upper bound on the globally closest alternative mean; we assign it no simple-hypothesis error probability or chi-squared calibration.

No proposed waveform has been applied to laboratory hardware.

4 Results

The baseline comparison completed all 99 cases. The first profile pass completed 297 of 300 jobs; a separate documented retry recovered the three failed original-model conductance profiles. Both passes are retained. None of the completed profiles found a minimum requiring a baseline refit under the recorded tolerance. The bootstrap completed all 96,000 planned refits, and the working OU likelihood comparison completed all 48 calibrations. Synthetic coverage, ambient/history and numerical sensitivity, and prediction uncertainty also completed their declared matrices and independent arithmetic audits. The parameter-design and model-discrimination matrices also completed their declared cases and independent audits.

4.1 Order and initialization

The startup assumption reverses the order ranking for the sine and joint calibrations (Fig. 1). For these two calibration sets, cold delay states give the lowest error at one stage among the compared formulations. Equilibrated states favor the pure-delay limit. The step experiment with equilibrated states gives the same error for every order, as expected from the delay's absence from its response. These comparisons concern calibration error under prescribed histories; they do not establish which history occurred before recording began. Table 1 gives the numerical comparison, including the step calibration, where the cold two-stage model improves on the cold one-stage fit. Table 2 gives the joint cold parameters.

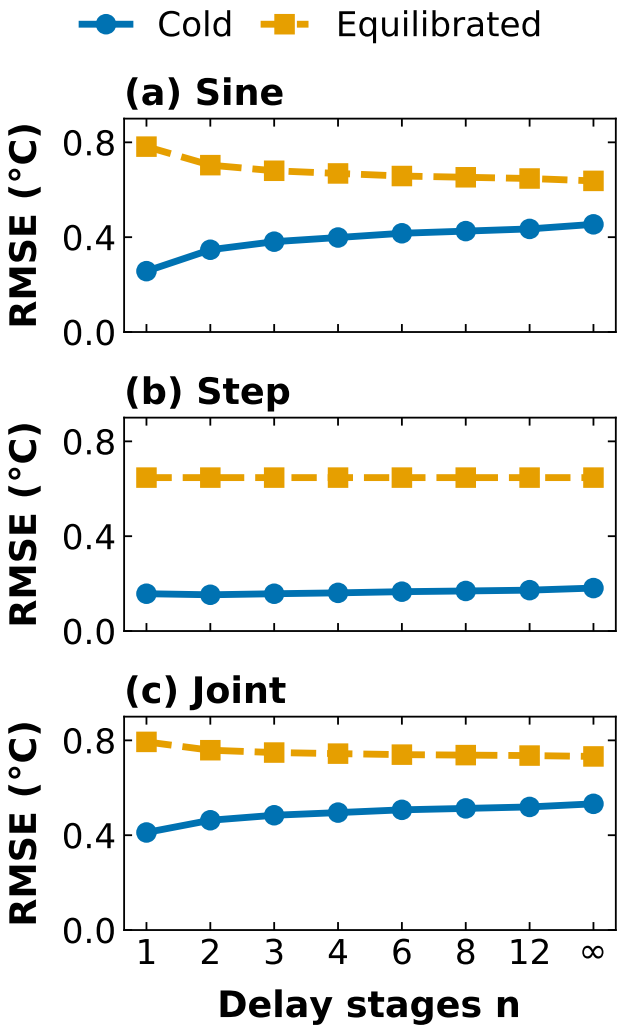


Figure 1: Initialization changes the fitted order ranking. Root mean squared temperature error (RMSE) for sine, step, and joint calibration is plotted for input–output parameter fits. Cold means $z_i(0) = 0$; equilibrated means $z_i(0) = u(0)$. The infinity label denotes the separately fitted pure-delay model, with the corresponding input prehistory. Lines connect categorical orders as guides. Every error excludes the first temperature used to initialize the model. No uncertainty intervals are implied by the markers.

Table 2: Joint cold-start estimates. Gain units are degrees Celsius per percentage-point command. Physical columns transform the IO estimates using fixed $c = 0.032$ W per percentage point; they are effective-model quantities. The one-stage row uses the representative $\tau \geq \theta$.

Model	K	τ (s)	θ (s)	U_a (W/K)	C_p (J/K)
$n = 1$	0.7744	150.697	16.555	0.04132	6.227
$n = 2$	0.7762	154.197	15.403	0.04123	6.357
$n = 3$	0.7768	155.291	15.074	0.04119	6.397
$n = 4$	0.7771	155.826	14.917	0.04118	6.417
$n = 6$	0.7774	156.353	14.764	0.04116	6.436
$n = 8$	0.7776	156.614	14.689	0.04115	6.445
$n = 12$	0.7777	156.874	14.614	0.04115	6.455
Pure delay	0.7780	157.391	14.466	0.04113	6.474

Table 3: Joint cold baseline fits evaluated without refitting on eleven duplicate-excluded validation files. RMSE values are in degrees Celsius; the held-out value pools all retained observations rather than averaging file RMSEs. File-mean share is $100 \sum_j N_j \bar{e}_j^2 / \sum_j \sum_i e_{ij}^2$, the retrospective algebraic contribution of each file's mean residual to pooled held-out error. It is not a causal attribution or a prospective bias correction.

Model	Calibration RMSE	Held-out RMSE	File-mean share (%)
$n = 1$	0.4115	3.3588	84.8
$n = 2$	0.4630	3.3608	83.4
$n = 12$	0.5192	3.3883	81.1
Pure delay	0.5323	3.4065	80.1

4.2 Temperature traces and residuals

The jointly calibrated cold models give closely spaced temperature predictions, while their residual traces show differences near startup and the sine peaks (Fig. 2). Residual structure persists over many successive observations. A visual match on the full temperature scale therefore does not establish independent sensor noise or reliable uncertainty coverage.

4.3 Transfer to supplied validation files

The four representative joint cold formulations give similar errors on the eleven duplicate-excluded validation files (Fig. 3). Differences among files are much larger than differences among these stage counts. Most pooled squared error is accounted for algebraically by the per-file mean residuals (Table 3). These offsets are retrospective diagnostics, without an identified physical cause or a prospective correction. Ambient, command history, heater conversion, thermal coefficients and omitted dynamics remain possible explanations. The supplied file names alone do not distinguish them.

4.4 Coordinate conditioning and physical ambiguity

The independently fitted physical and input–output formulations reach sums of squares that agree to numerical precision in the joint cold cases (Table 5). Reexpressing the same sensitivity matrix changes its condition number. Moment coordinates do not improve this particular metric at these fitted points. These differences concern the coordinate metric; they do not distinguish observationally

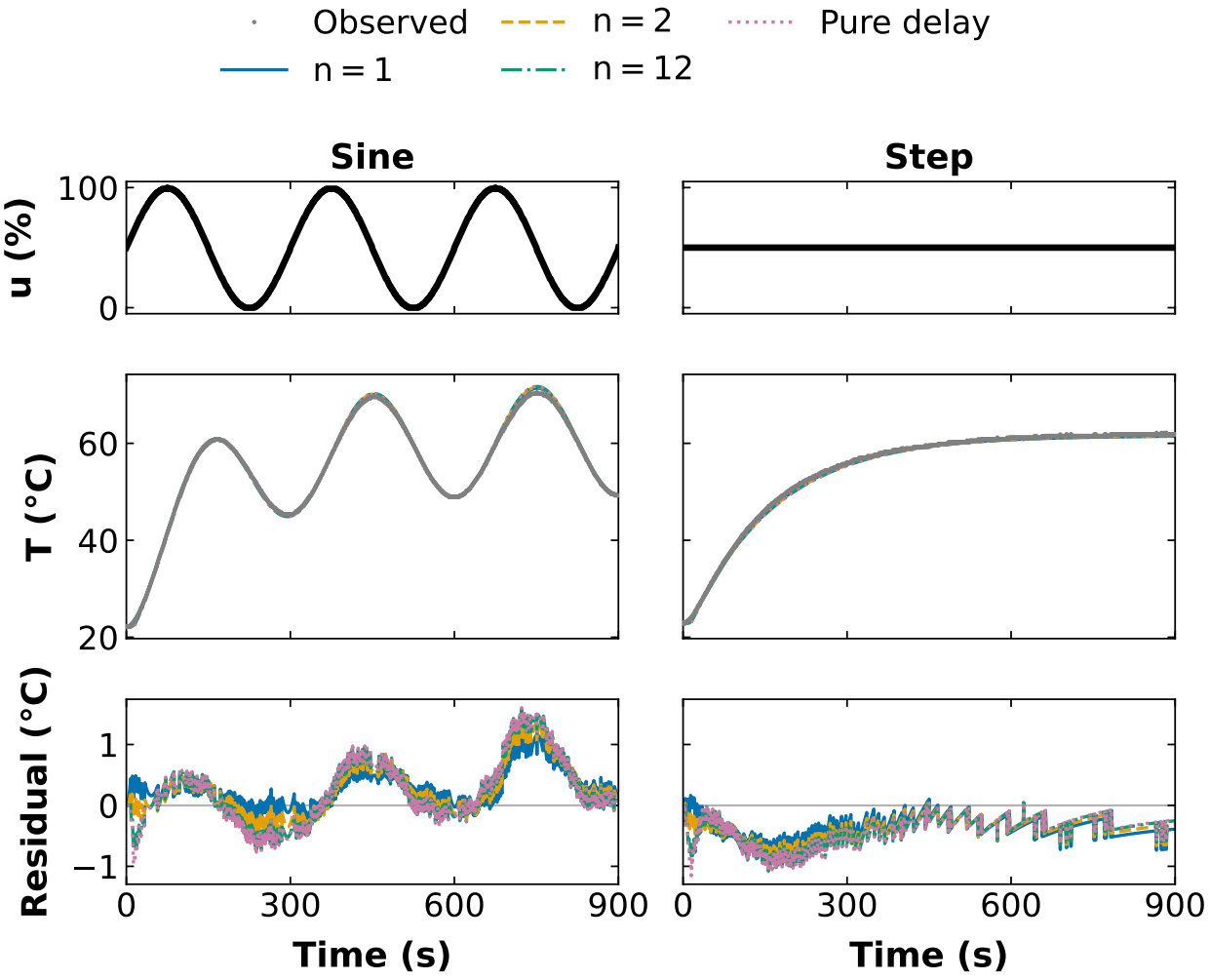


Figure 2: The residual scale exposes differences hidden in the temperature overlays. Columns show the sine and step experiments; rows show heater input u, measured temperature T with jointly fitted cold-start mean predictions, and residuals defined as prediction minus observation. All recorded calibration samples are plotted. The thermal initial condition and assumed ambient equal the first observation, and the delay prehistory is zero. The four curves use independent joint fits for their respective formulations. These are mean predictions; uncertainty bands are not shown.

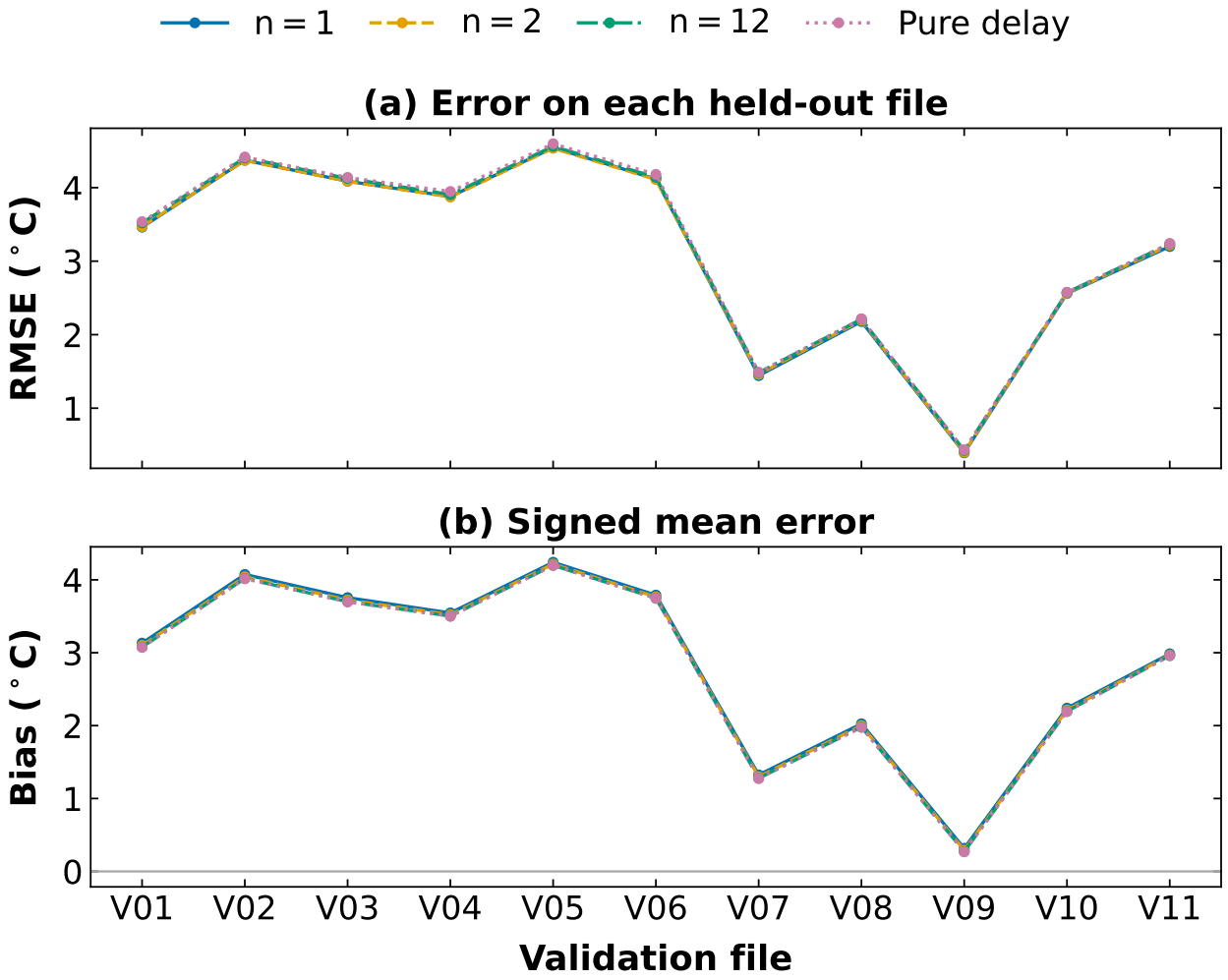


Figure 3: Transfer error varies more among supplied files than among the representative cold delay formulations. Panels show RMSE and signed mean residual, defined as prediction minus observation, excluding each initial temperature. Parameters come from the two joint calibration records and are not refitted on these files. Codes are defined in Table 4; lines join categorical file codes as guides. Each prediction uses the file's first temperature as both its thermal initial condition and assumed ambient. Markers do not imply independent replication, and no uncertainty interval is shown.

Table 4: Validation-file codes. Restore the common prefix `validation_experiment_` to each supplied filename. N excludes the initial temperature, which also supplies the assumed ambient for that file; it is not an independent ambient measurement. File labels do not establish independent replication or a change in physical environment.

Code	Supplied filename suffix	Duration (s)	T_0 (°C)	N
V01	`env_2_sin_2_50_run_1.csv`	900.0	20.91	899
V02	`env_2_sin_2_50_run_2.csv`	900.0	20.26	900
V03	`env_2_sin_2_50_run_3.csv`	900.0	20.26	900
V04	`env_2_sin_50_50_run_1.csv`	900.0	19.94	900
V05	`env_2_sin_50_50_run_2.csv`	900.0	20.58	898
V06	`env_2_sin_50_50_run_3.csv`	900.0	20.26	898
V07	`env_2_sin_5_50_run_2.csv`	900.0	23.48	900
V08	`env_2_sin_5_50_run_3.csv`	900.0	24.13	900
V09	`env_2_step_50_run_2.csv`	900.0	24.13	900
V10	`env_2_step_50_run_3.csv`	900.0	23.81	900
V11	`sin_5_50.csv`	1000.0	24.77	989

Table 5: Coordinate metrics for joint cold fits. The first three columns are sensitivity condition numbers at the same IO model point, transforming the information matrix without refitting. Moment coordinates are $(\log K, \log m, \ell)$ with $m = \tau + \theta$ and $\ell = \log[\theta/\tau]$. The last column compares independently fitted IO and physical sums of squares, in squared degrees Celsius. Conditioning changes do not imply additional information.

Model	IO logs	Physical logs	Moment	$\lvert\Delta S\rvert$
$n = 1$	23.1	19.38	27.76	6.3×10^{-12}
$n = 2$	21.04	17.56	24.82	1.5×10^{-11}
$n = 3$	20.61	17.19	24.15	1.3×10^{-12}
$n = 4$	20.44	17.04	23.86	4.3×10^{-11}
$n = 6$	20.29	16.9	23.6	2.1×10^{-11}
$n = 8$	20.23	16.84	23.48	3.2×10^{-12}
$n = 12$	20.16	16.78	23.36	3.4×10^{-13}
Pure delay	20	16.64	23.09	8.0×10^{-11}

equivalent model explanations. Table 6 shows the corresponding local parameter correlations. Transforming the capacity combines gain and time-constant uncertainty, so the physical-coordinate correlations need not resemble the IO correlations even though the predicted thermal response is unchanged.

The positive information spectra are similar across the reduced cold models, while their native local dimensions differ from the original model and the equilibrated step case (Fig. 4). The weakest local direction is dominated by relative delay changes at these joint cold points, with smaller compensating time-constant changes. Its components change after reexpressing the same model in physical coordinates (Fig. 5). Neither local eigenanalysis detects the one-stage pole-label exchange by itself.

The original model has a stronger ambiguity. For each calibration, the saved structural calculation sampled 403 positive physical parameter combinations with the same transfer function. Their largest prediction difference was below 1.8×10^{-12} °C. Conductance U_{a} remained fixed while U_{b}, $C_{\mathrm{p,H}}$ and $C_{\mathrm{p,S}}$ varied. Appendix A gives the analytic family and its positivity limits; sampled ranges are not

Table 6: Local IID correlations at the same joint cold model point. IO coordinates are $(\log K, \log \tau, \log \theta)$; physical coordinates are $(\log U_{\mathrm{a}}, \log C_{\mathrm{p}}, \log \theta)$. The physical covariance is transformed without refitting. Correlation changes reflect the chosen quantities, not extra information. One-stage entries describe the local canonical pole mode, excluding its exchanged global label.

Model and log coordinates	ρ_{12}	ρ_{13}	ρ_{23}
$n = 1$, IO	0.682	-0.366	-0.696
$n = 1$, Physical	-0.535	0.366	-0.711
$n = 2$, IO	0.678	-0.312	-0.602
$n = 2$, Physical	-0.511	0.312	-0.616
$n = 12$, IO	0.676	-0.269	-0.523
$n = 12$, Physical	-0.494	0.269	-0.535
Pure delay, IO	0.675	-0.260	-0.506
Pure delay, Physical	-0.491	0.260	-0.518

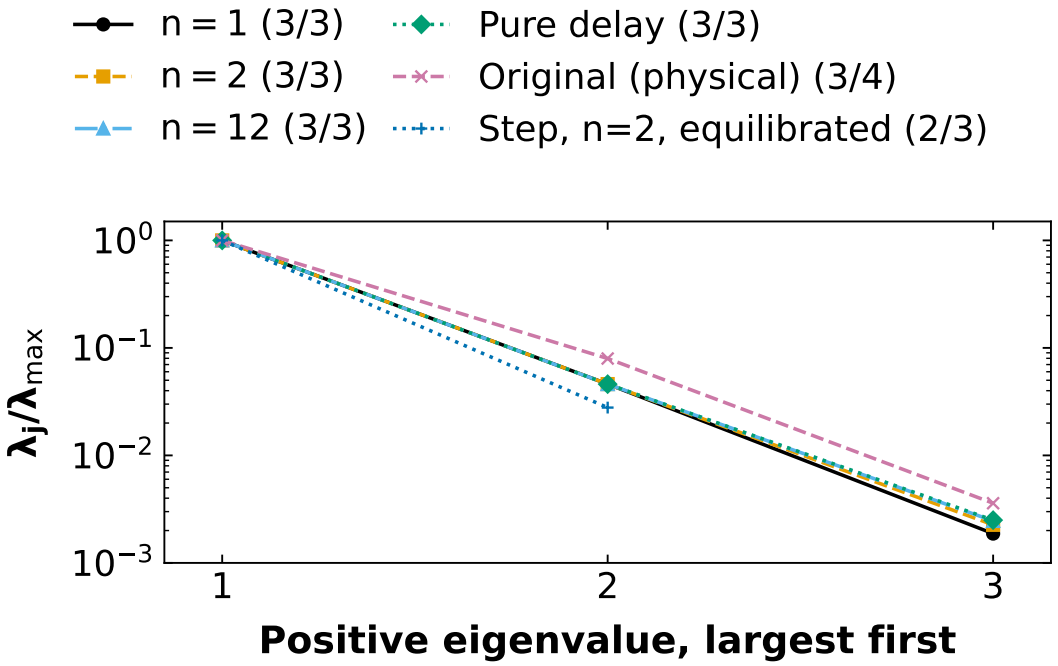


Figure 4: Positive information eigenvalues and structural null directions must be counted separately. Values above the relative sensitivity-rank cutoff are normalized by each case's largest eigenvalue. Legend parentheses give local identifiable rank divided by native parameter dimension. The four reduced curves use joint cold fits in log input–output coordinates and overlap closely. The original joint model uses four physical logs; the step comparison uses two-stage equilibrated delay states. Their null eigenvalues are omitted rather than assigned a plotting floor. These coordinate-dependent spectra are not an invariant ranking of information magnitude between formulations. Lines join ordered eigenvalues as guides.

Table 7: Identifiable boundary functions for the full strictly positive original-model family, propagated from the joint cold one-stage transfer fit. Intervals are conditional 95 percent percentiles from 400 saved draws per method, before numerical physical-parameter boxes. The heater-capacity infimum and supremum are open family endpoints; the sensor-capacity and coupling maxima are attained. These are not marginal intervals for an arbitrary physical parameter.

Boundary function	Fitted value	IID simulation	Circular block 60
$\inf C_{\mathrm{p,H}}$ (J/K)	0.684	$[0.6792, 0.6896]$	$[0.6448, 0.7276]$
$\sup C_{\mathrm{p,H}}$ (J/K)	6.227	$[6.214, 6.238]$	$[6.144, 6.315]$
$\sup C_{\mathrm{p,S}}$ (J/K)	2.783	$[2.765, 2.799]$	$[2.647, 2.91]$
$\sup U_{\mathrm{b}}$ (W/K)	0.07451	$[0.07359, 0.07528]$	$[0.06801, 0.08156]$

confidence intervals.

Figure 6 shows the joint family and the forward numerical agreement along it. A selected optimizer point is one compatible member. By contrast, the family's limiting capacities and maximum coupling are functions of identifiable transfer coefficients. Table 7 and Fig. 7 propagate saved one-stage bootstrap draws to these boundary functions. Larger residual blocks widen their conditional percentile intervals. This quantifies uncertainty in the family limits without assigning an arbitrary member the status of a physical estimate.

The analytic family also exposed an interpolation failure in the first-pass profiles. All nine redundant physical-parameter grid sets excluded some known equivalent members. Their sampled objectives were correct, but interpolation across the rising edges gave misleading endpoints. A separate refinement added verified equivalence witnesses and bisected the observed threshold crossings. The selected catalog now includes the complete analytically compatible intervals inside the numerical fitting box for all nine profiles. First-pass files remain available, and the revised numerical endpoints do not make the individual physical parameters identifiable. The same selected refinement treats the cold one-stage delay profiles in both coordinates and all three calibration sets, and the representative joint cold two-stage, twelve-stage and pure-delay profiles. Its arithmetic, newly evaluated forward objectives and family inclusion passed a separate audit. These checks do not certify unobserved modes or global nuisance minima.

4.5 Conditional delay uncertainty

Table 8 compares delay uncertainty for representative joint cold formulations. Increasing the residual block length widens the percentile intervals relative to independent Gaussian simulation. The one-stage profile has two accepted pole-label modes; canonical bootstrap summaries refer to only the representative with $\tau \geq \theta$. A finite full-coordinate interval is withheld where the recorded experiment leaves delay unobservable. The table compares assumptions rather than estimating a single unconditional physical uncertainty.

Figure 8 displays the two one-stage profile modes on separate delay windows. Figure 9 compares the conditional intervals across residual block lengths. The apparent precision changes with the residual assumptions even when the fitted thermal trajectories remain close.

4.6 Synthetic coverage exposes sensitivity to serial dependence

All 3600 prespecified synthetic datasets finished without failed draws. Each of the 84 delay-method comparisons returned an interval for every requested dataset, so its method-defined and unconditional

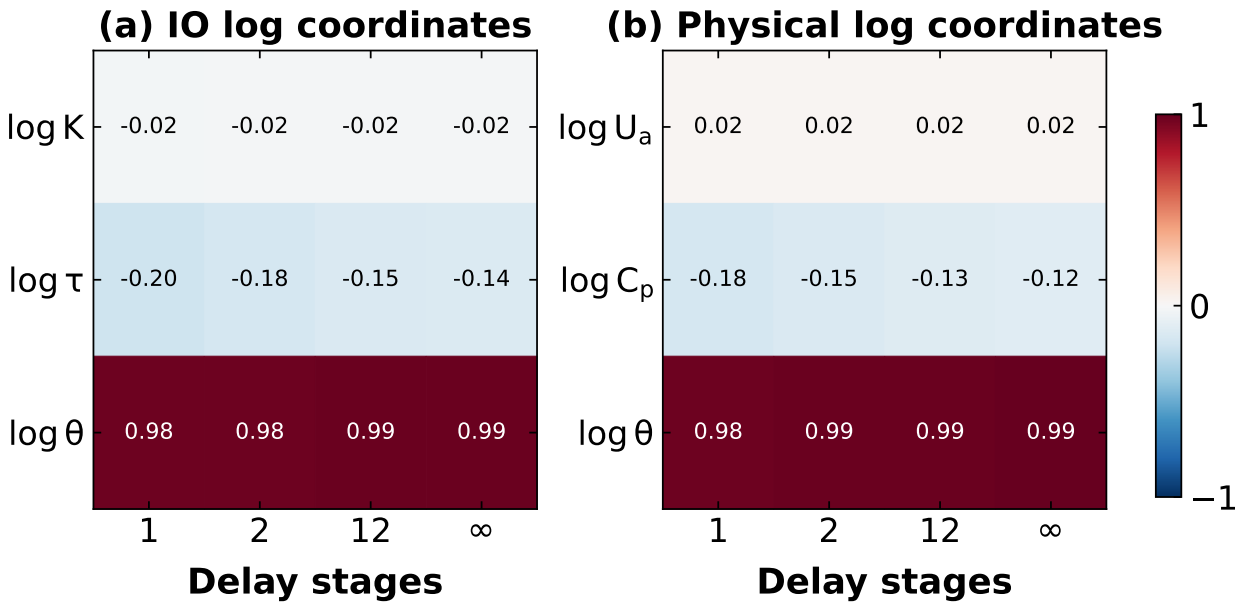


Figure 5: Weakest local information directions depend on the coordinate metric. Columns give joint cold fits for the indicated stage count, with infinity denoting pure delay. Entries are components of unit eigenvectors for the smallest information eigenvalue. The largest absolute component is assigned positive sign. Physical-coordinate directions are re-diagonalized after the exact transformation at the same fitted point; they are not additional observations. The displayed directions concern one local pole representative and do not remove global parameter ambiguities.

Table 8: Joint cold delay uncertainty in seconds. Profile sets use a profiled common IID variance and retain both one-stage pole modes. Percentile intervals use 400 Gaussian simulations at fixed $\sigma = 0.25$ degrees Celsius or 400 circular residual-block refits, respectively. Bootstrap one-stage delays use the canonical pole representative. OU entries are local standard deviations with variance profiled and length included in the Hessian, not interval endpoints. These procedures make different assumptions and are not interchangeable coverage guarantees.

Model	IID profile set	IID simulation	Circular block 60	OU local SD
$n = 1$	$[16.35, 16.76] \cup [150.2, 151.2]$	$[16.44, 16.69]$	$[15.62, 17.56]$	0.585
$n = 2$	$[15.2, 15.6]$	$[15.3, 15.51]$	$[14.31, 16.44]$	0.606
$n = 12$	$[14.41, 14.82]$	$[14.52, 14.72]$	$[13.25, 15.98]$	0.654
Pure delay	$[14.26, 14.67]$	$[14.38, 14.56]$	$[13.34, 15.68]$	0.317

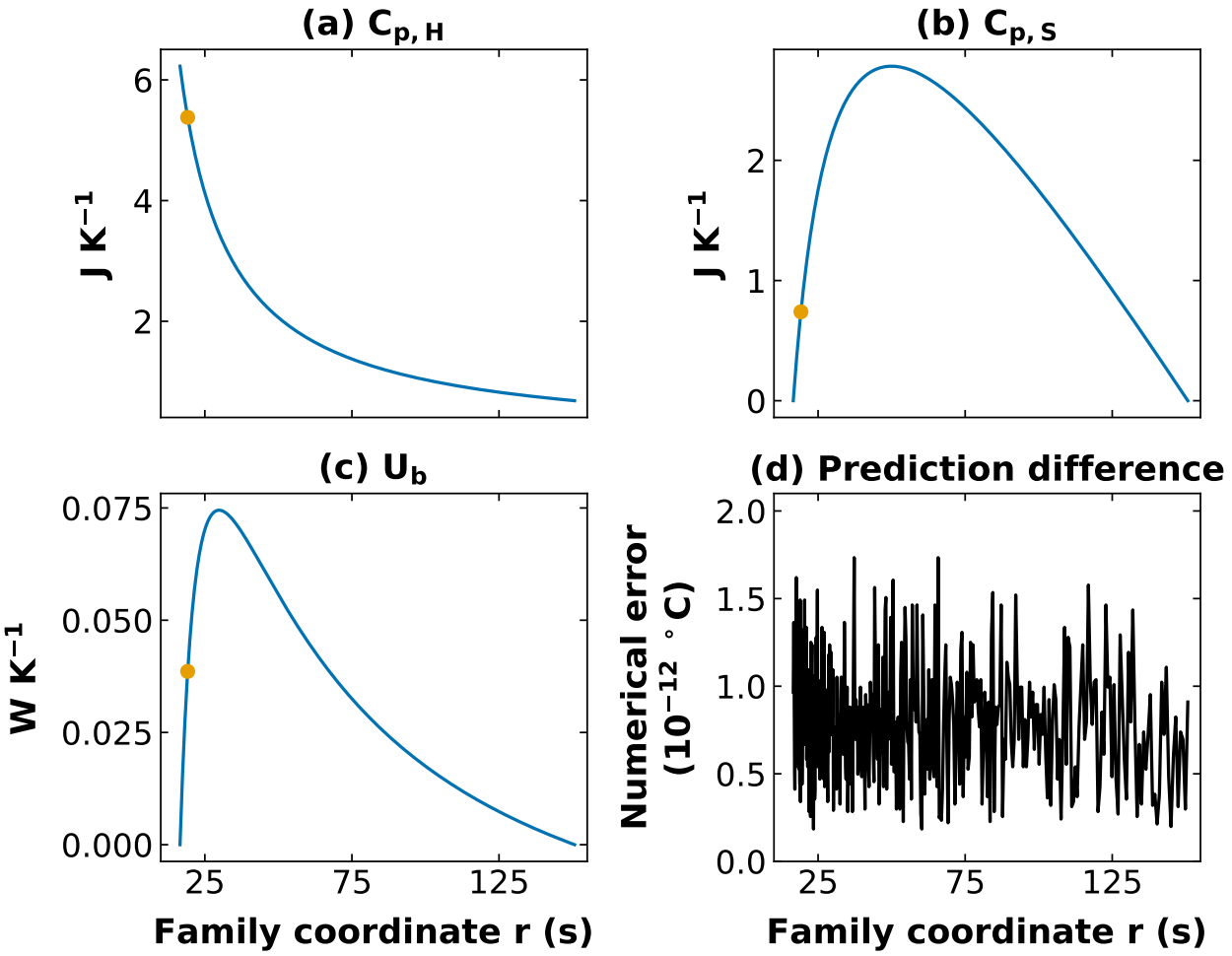


Figure 6: Different original physical parameters give the same measured temperature response. The joint cold transfer function generates 403 positive members parameterized by $r = C_{\mathrm{p,S}}/U_{\mathrm{b}}$. Panels (a)–(c) show the heater capacity, sensor capacity and coupling conductance. Orange dots mark the member selected by the original optimizer. Panel (d) shows maximum absolute prediction differences over both calibration records, exposing floating-point agreement rather than measurement precision. Of the displayed members, 401 fall inside the original numerical fitting box. Strictly positive family endpoints are approached but not attained; the curves are not confidence regions.

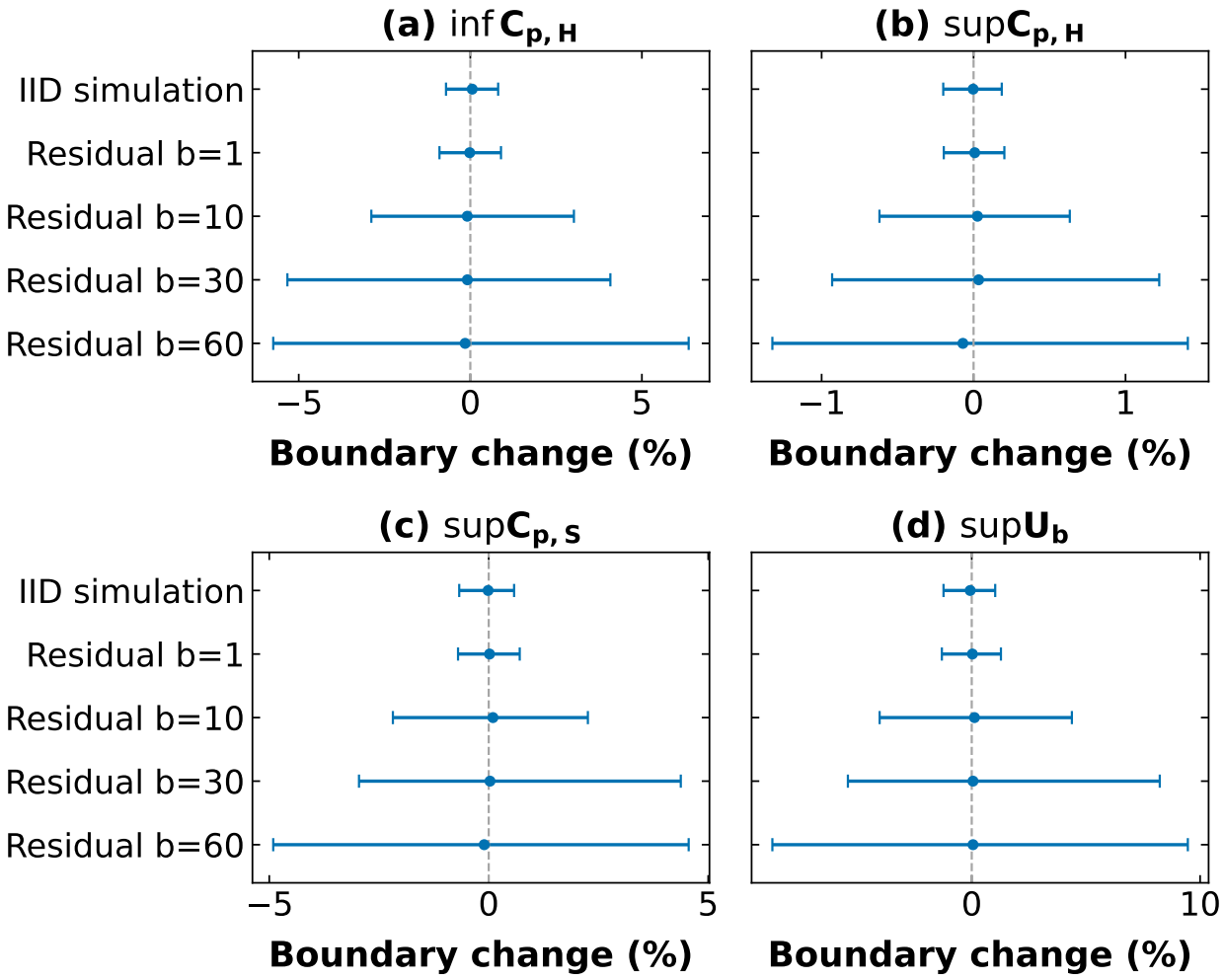


Figure 7: Uncertainty in identifiable physical-family boundaries can be propagated without selecting a physical family member. Panels show the heater-capacity infimum and supremum, sensor-capacity supremum and coupling supremum. Each bar is a conditional 95% percentile interval from 400 saved joint cold one-stage bootstrap draws, expressed as percent change from its fitted boundary value; dots mark medians. The strictly positive family precedes rectangular numerical fit boxes. These are intervals for boundary functions, excluding heater-calibration and model-discrepancy uncertainty; they are not marginal intervals for $C_{\mathrm{p,H}}$, $C_{\mathrm{p,S}}$ or U_{b}.

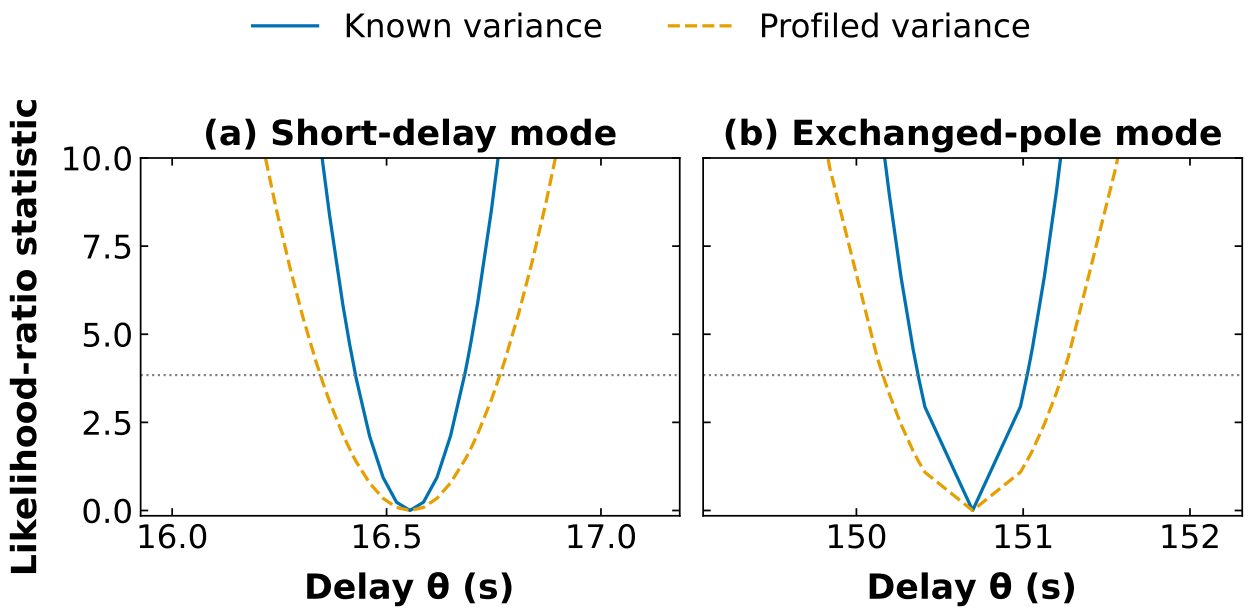


Figure 8: The one-stage cold model has two distinct delay-label modes. Each panel shows a separate declared window of the joint delay profile; the intervening parameter range is omitted. The blue solid curve uses fixed hypothetical sensor standard deviation 0.25 °C, and the orange dashed curve profiles a common IID variance. Lines connect saved nuisance-optimized grid values. The dotted line is the nominal 95% chi-squared threshold with one degree of freedom. Negative likelihood-ratio roundoff is clamped to zero for display. Neither regular threshold assumptions nor real-data coverage follow from the profile shape.

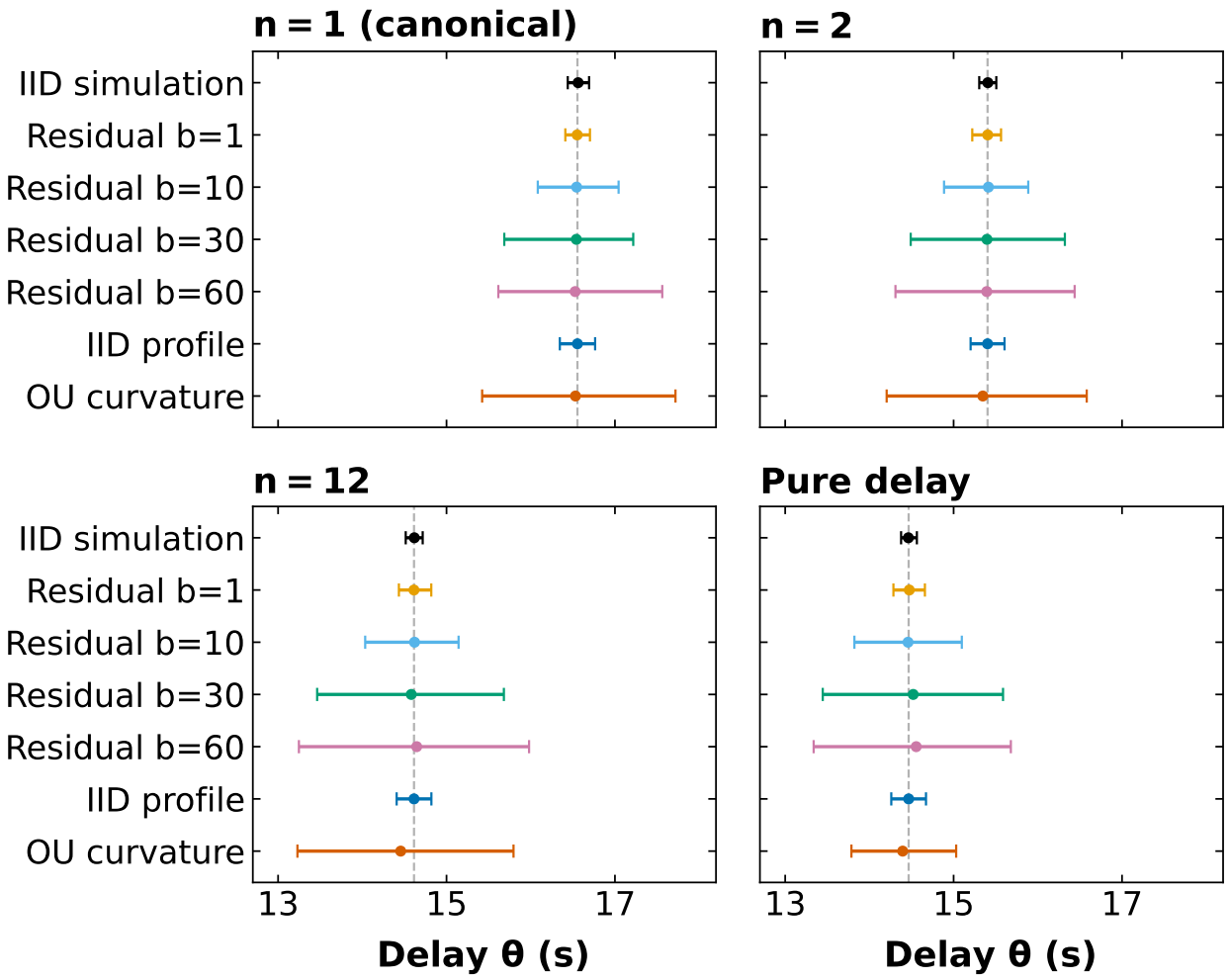


Figure 9: Residual assumptions change conditional delay intervals. All panels use joint cold fits; the one-stage panel displays the canonical short-delay representative. Bootstrap bars are percentile intervals from 400 saved draws per method, with dots at medians. The profile bar uses the unknown-variance IID accepted component containing the baseline estimate. OU bars exponentiate local normal log-delay endpoints from the observed mean/length Hessian with variance profiled. Their dots are fitted OU delays. Gray vertical lines mark baseline least-squares delays. Residual blocks count retained samples. These are nominal conditional intervals under different assumptions, not a common unconditional physical uncertainty or simultaneous confidence statement.

Table 9: Delay-interval coverage counts for four selected procedures. Each entry is covered/defined, from 300 requested datasets at that generating point and true covariance; – would denote no defined interval. All intervals have nominal 95% reference levels. IID known/fitted refer to Gaussian log-Wald intervals; HAC60 uses a Bartlett bandwidth of 60 retained samples; profile fitted uses a common profiled IID variance and retains disconnected components. OU10/OU60 are true correlation lengths in seconds, with fixed marginal standard deviation. Full method-specific failure bounds and grid diagnostics are retained in the coverage CSV; Wilson Monte Carlo intervals and all seven methods appear in Fig. 10.

Model	Error law	IID known	IID fitted	HAC60	Profile fitted
$n = 1$	IID	278/300	278/300	273/300	278/300
$n = 1$	OU10	105/300	103/300	263/300	102/300
$n = 1$	OU60	73/300	66/300	235/300	66/300
$n = 2$	IID	281/300	280/300	272/300	280/300
$n = 2$	OU10	110/300	110/300	266/300	108/300
$n = 2$	OU60	61/300	56/300	249/300	55/300
$n = 12$	IID	282/300	281/300	276/300	280/300
$n = 12$	OU10	106/300	104/300	262/300	104/300
$n = 12$	OU60	59/300	53/300	248/300	53/300
Pure delay	IID	290/300	288/300	283/300	288/300
Pure delay	OU10	106/300	104/300	273/300	103/300
Pure delay	OU60	74/300	68/300	254/300	68/300

coverage rates coincide. Table 9 gives exact counts for four procedures; Fig. 10 includes all seven and their Monte Carlo uncertainty.

Under IID Gaussian generation, the IID local and profile procedures gave coverage near the nominal reference. Applying those same procedures under OU generation produced severe undercoverage, especially at the longer correlation length. Estimating a common variance did not compensate for ignoring serial covariance. HAC intervals improved coverage under correlated generation, with substantial bandwidth sensitivity. Even the largest tested bandwidth fell below the nominal reference in the OU60 comparisons. The synthetic thermal mean is correct by construction here; these failures do not require between-record mean discrepancy.

The profile comparison retains a numerical limitation. Three-point quadratic interpolation changed 68 of 300 coverage classifications for the two-stage known-variance profile under OU60 generation. This is a sensitivity check on the saved grid, without new nuisance optimization or an endpoint-accuracy certificate. Both primary and secondary classifications remain conditional diagnostics of an IID likelihood applied to correlated errors. The full CSV retains this diagnostic for every profile case; interpreting the known-variance profile rates as exact continuous-profile coverage would overstate their numerical verification.

Figure 11 compares median interval widths. The one-stage profile width sums its two accepted pole-label components rather than measuring the gap between them. A wider accepted set can therefore concern a different global parameter representation from a local interval around one pole label. These simulations fix four generating points, the recorded inputs/times and the initial-temperature, ambient and heater-calibration assumptions. They do not measure coverage of the real-data block-bootstrap intervals or uncertainty from the observed between-record discrepancy.

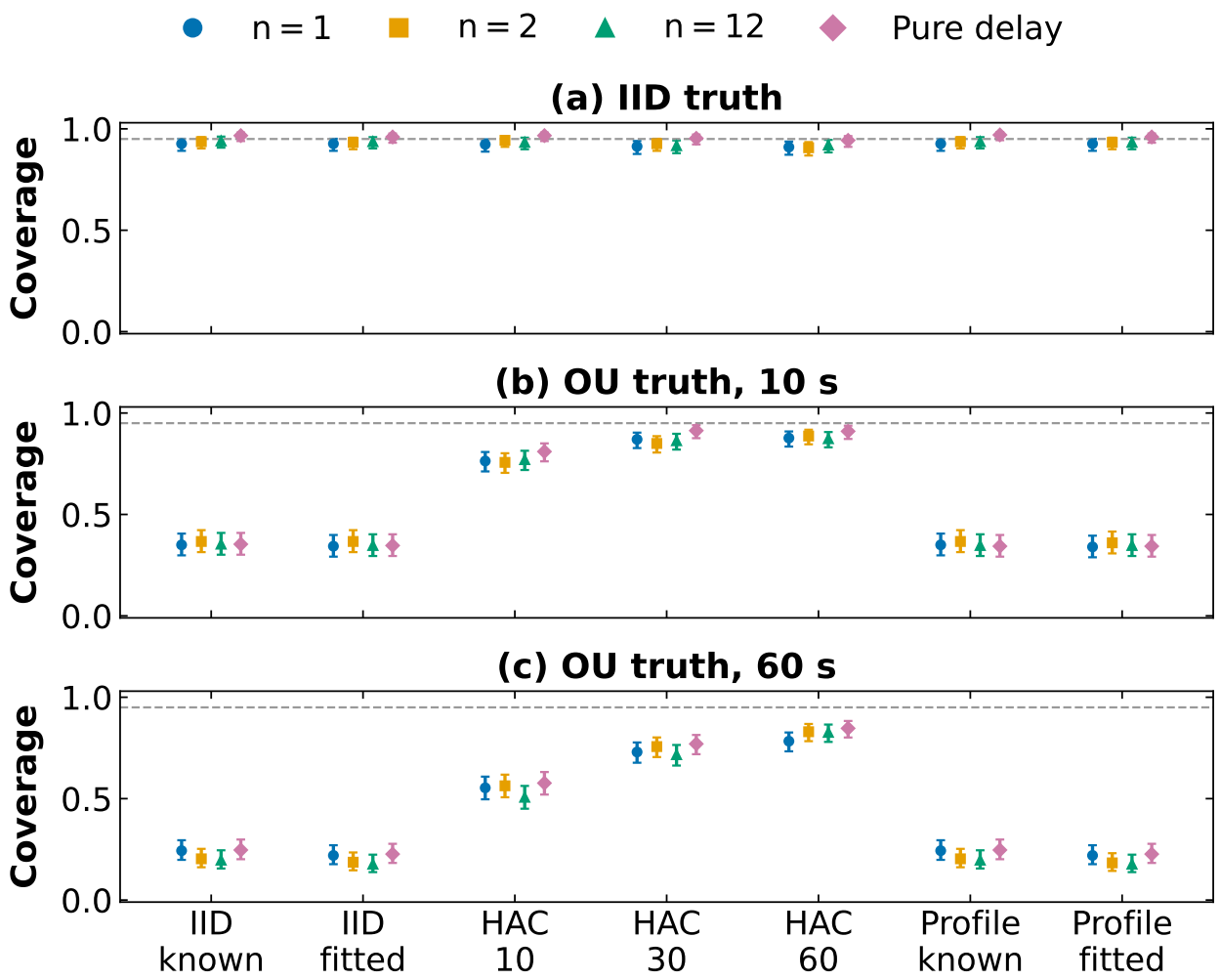


Figure 10: Serial dependence reduces coverage of IID-based delay intervals in controlled simulation. Each point is covered/defined over 300 requested datasets for one model and true error process; bars are pointwise 95% Wilson intervals for Monte Carlo sampling uncertainty. All displayed procedures returned intervals for every dataset. The dashed line is nominal 0.95 coverage. Known/fitted denote fixed or profiled IID variance; local IID intervals are Gaussian log-Wald intervals and profiles retain all accepted components. HAC labels give Bartlett bandwidths in retained samples. All true errors have the same marginal standard deviation; rows change their correlation law. The thermal mean and startup are correctly specified in each generating scenario. Profile sets retain the declared finite-grid limitation. These are generating-point results, without a universal or real-discrepancy coverage guarantee.

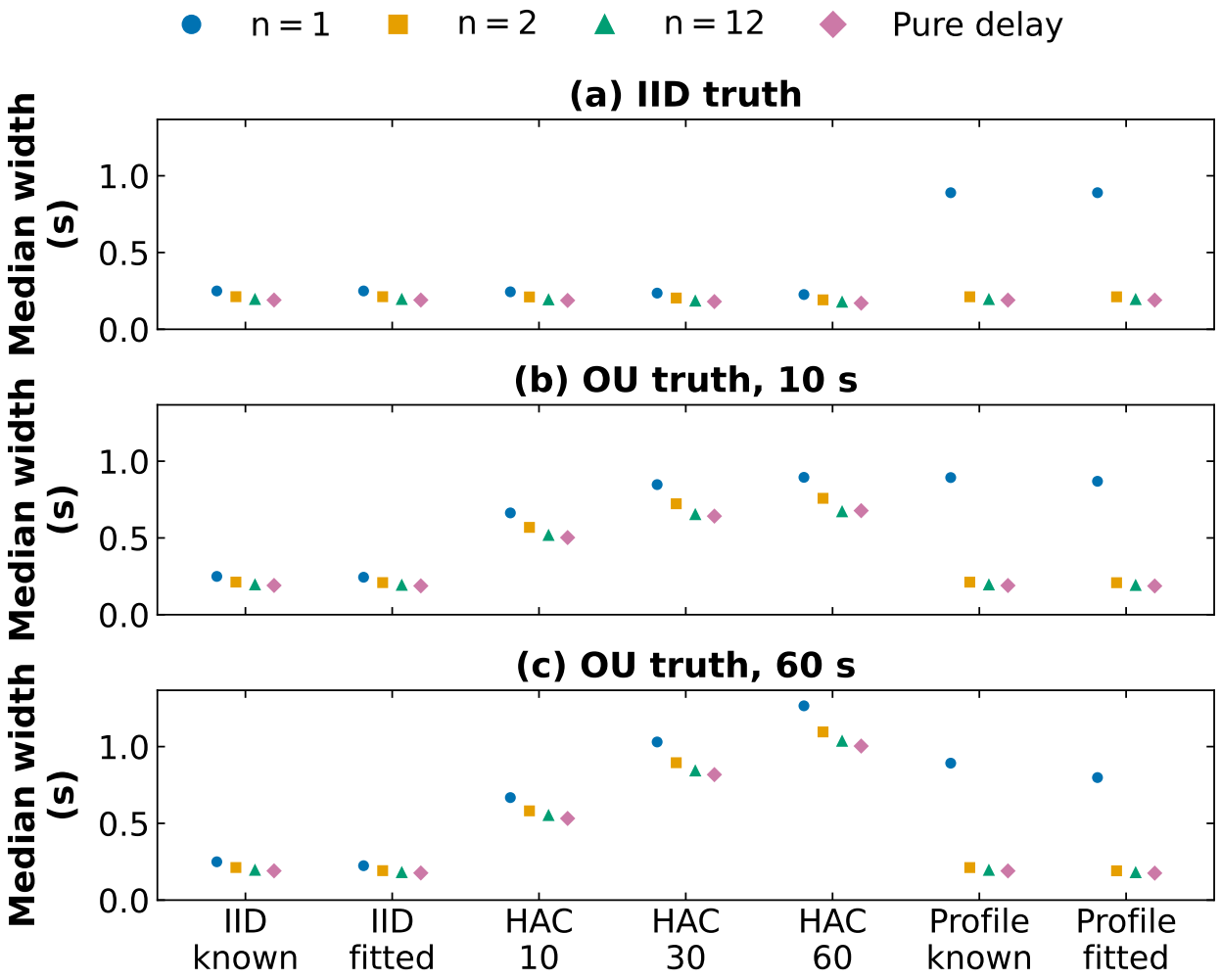


Figure 11: Median delay-interval widths depend on the uncertainty procedure and true serial dependence. Models, methods and simulation denominators match Fig. 10. All panels share the same width scale. Profile widths sum accepted component lengths, excluding intervening gaps; the one-stage profile includes both pole-label modes. These width summaries do not establish nominal coverage. Exact counts, width distributions, exploration flags and interpolation sensitivity remain in the saved coverage summary.

Table 10: Working OU fits to the two joint cold records. The scale is the maximum-likelihood stationary residual standard deviation. ACF1 is the centered sample lag-one correlation of filtered innovations. Strong remaining negative correlation challenges this one-component residual covariance; a high fitted likelihood does not validate it.

Model	Length (s)	$\hat{\sigma}$ (°C)	ACF1, sine	ACF1, step
$n = 1$	17.46	0.4088	-0.5333	-0.492
$n = 2$	22.14	0.459	-0.5291	-0.4935
$n = 12$	27.91	0.5151	-0.5253	-0.4877
Pure delay	29.33	0.5289	-0.518	-0.4811

4.7 Nonregular curvature and the delay-chain limit

Two deterministic checks distinguish local singularity from exact physical ambiguity (Fig. 12). Near a repeated pole, the squared temperature distance grows quartically along the fixed-mean perturbation, consistent with the canceled first derivative in Appendix A. The rank-deficient quadratic approximation misses this higher-order distinction. Separately, the cold constant-step onset excludes a uniform $O(n^{-1})$ approximation error bound. At $t = \theta$, the pure-delay response has not risen, whereas the finite-chain rise decays as $n^{-1/2}$ with the derived coefficient. These are fixed-parameter model implications, rather than comparisons of refitted TC-Lab errors.

4.8 The working correlated-noise model remains inadequate

The independent innovations audit reproduced every saved OU likelihood within 2.6×10^{-12}. This validates likelihood arithmetic, not the residual model. Filtered innovations retain strong negative lag-one correlation (Table 10). The fitted length therefore cannot be treated as a measured thermal disturbance time scale. Pooled held-out prediction error also remains much larger than calibration error. Allowing serial covariance changes conditional uncertainty without establishing that the thermal mean transfers between records. Figure 13 compares raw residuals and filtered innovations within each OU fit. For reference, filtering an IID component at regular sampling gives interior lag-one correlation $-\phi/(1 + \phi^2)$, approaching $-1/2$ for ϕ near one (Appendix B). This provides one explanation to investigate, without identifying a unique error mechanism from the observed negative correlation.

4.9 Ambient and history assumptions shift the fitted delay

Allowing record-specific ambient offsets reduces joint calibration error more than allowing only an initial history fraction in the representative models (Fig. 14). The fitted delay also changes, although the cold one-stage formulation retains the lowest calibration error among these four representatives. With both nuisance terms free, every displayed step-record ambient offset reaches the imposed upper bound (Table 11). The lower residual error therefore does not identify an independently measured ambient temperature. Extending that bound until the fit looks satisfactory would change the sensitivity question rather than resolve the missing measurement.

The nuisance fit retains the first sensor observation as its thermal initial condition while changing the assumed ambient and delay-state history. Its history fraction describes a declared state family; it does not reconstruct an unrecorded heater command. The separate constant-input derivation in Appendix B explains why a fitted history or ambient can also introduce exact parameter ambiguity. A favorable joint residual score does not establish estimability in every experiment.

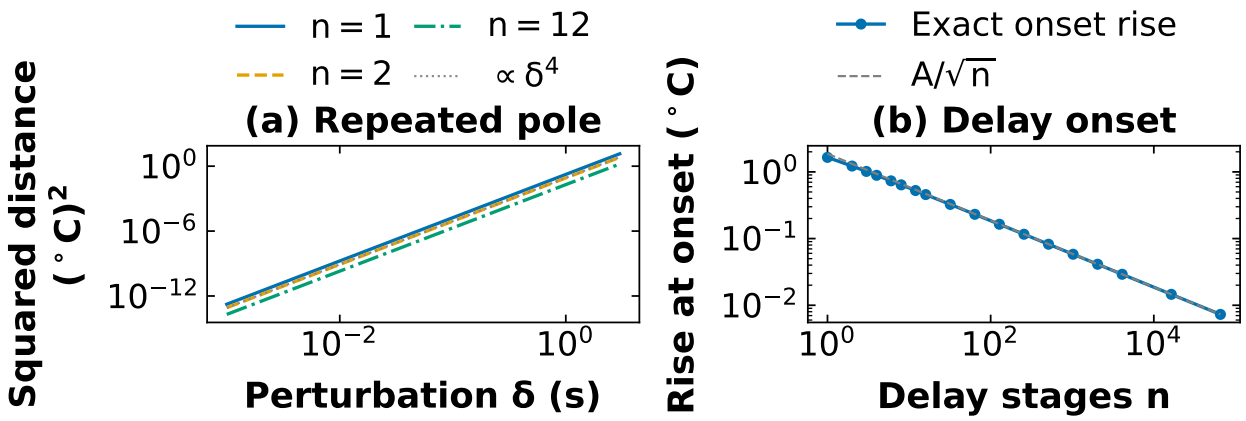


Figure 12: Two deterministic limits challenge routine local approximations. (a) Squared prediction distance over the recorded sine command for $K = 0.78$, $\tau = 10 + \delta$ and $\theta = 10n - \delta$, with cold states. The dotted quartic guide is anchored at the smallest positive two-stage perturbation. (b) Exact cold constant-step temperature rise at $t = \theta$ for $K = 0.78$, $\tau = 150$ s, $\theta = 18$ s and $u = 50$ percentage points. The reference coefficient is $A = Ku\theta/(\tau\sqrt{2\pi})$; the pure-delay rise is zero at that instant. The calculations use separate parameter and input settings and do not constitute fitted order selection or uncertainty coverage.

Table 11: Joint fits with free ambient and history. Ambient offsets are in degrees Celsius; history fractions γ multiply the initial command. Gain units are degrees Celsius per command percentage point. Every displayed step offset reaches the imposed +2 degree bound. These are sensitivity fits under declared bounds, not independently measured ambient temperatures or histories.

Model	K (°C/%)	θ (s)	$\Delta T_{a,\mathrm{sin}}$ (°C)	$\Delta T_{a,\mathrm{step}}$ (°C)	γ_{sin}	γ_{step}
$n = 1$	0.7390	17.159	1.339	2.000	0.000	0.004
$n = 2$	0.7406	16.350	1.336	2.000	0.088	0.089
$n = 12$	0.7421	15.932	1.321	2.000	0.195	0.173
Pure delay	0.7424	15.867	1.318	2.000	0.218	0.192

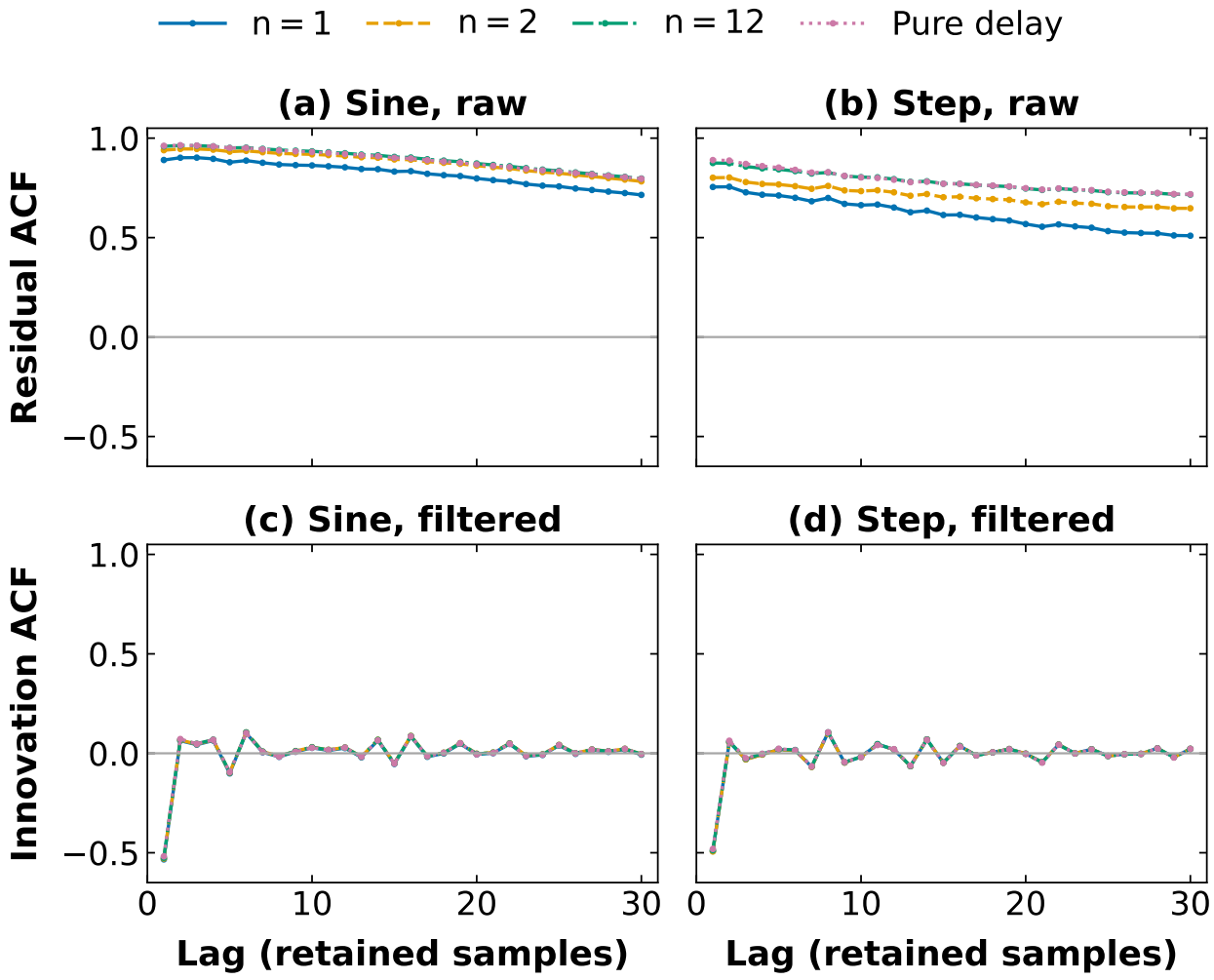


Figure 13: A one-component OU covariance removes much low-frequency residual correlation while leaving strongly negative first-lag innovations. Columns show calibration records; rows show centered sample autocorrelations before and after the stationary OU filter. Each curve uses its own jointly fitted cold thermal mean and OU length in both rows, isolating the filter comparison within that fit. Lag zero is omitted. These are empirical residual diagnostics, without nominal whiteness-test bands or a unique physical interpretation of the residual process.

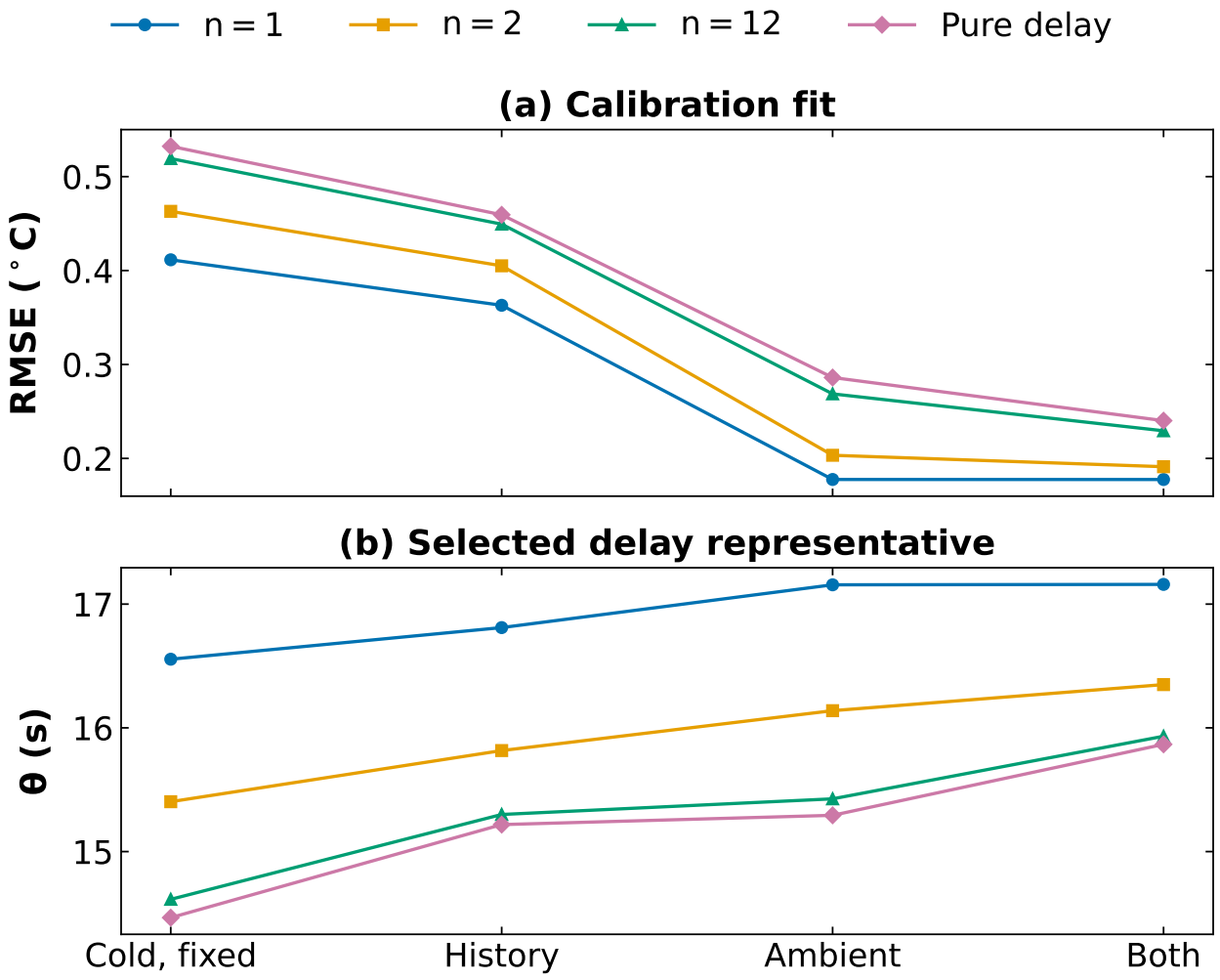


Figure 14: Ambient and initial-history assumptions change joint calibration error and the selected delay. Cold, fixed is the original linear-hold fit; History frees one delay-state history fraction per record with ambient fixed; Ambient frees one ambient offset per record with cold delay states; Both frees both. Offset bounds are $\pm 2\,^{\circ}\mathrm{C}$ and history fractions lie in $[0, 1]$. The two thermal experiments share their model parameters. Lines join categorical assumptions; markers are fitted representatives without confidence intervals. The one-stage delay uses the selected short-delay pole label.

Table 12: Median pointwise mean-band widths and descriptive V01 containment by nominal observation-prediction bands. The last column uses observation bands with an added fitted IID or OU marginal future-error variance; it is not containment by the narrower mean bands. V01 is defined in Table 4. These fractions on one observed record are not repeated-experiment coverage estimates. HAC and block-residual methods supply no future-error law here and therefore no observation-band containment.

Model, method	Sine width (°C)	V01 width (°C)	V01 obs. fraction
$n = 1$, IID fitted	0.078	0.061	0.135
$n = 1$, OU fitted	0.439	0.347	0.139
$n = 2$, IID fitted	0.088	0.077	0.140
$n = 2$, OU fitted	0.544	0.479	0.154
$n = 12$, IID fitted	0.099	0.094	0.150
$n = 12$, OU fitted	0.662	0.629	0.166
Pure delay, IID fitted	0.102	0.097	0.161
Pure delay, OU fitted	0.560	0.512	0.171

4.10 Command reconstruction changes the inferred delay

Changing the command hold shifts the fitted delay while producing smaller changes in calibration RMSE (Fig. 15). A backward-Euler fit can even reduce the residual score relative to exact propagation. That does not make its discretization a more accurate description of the continuous thermal system: fitted parameters can absorb numerical and command-reconstruction differences. The recorded time/command pairs do not identify the command waveform between samples. Consequently, the narrowest conditional delay intervals should not be interpreted as resolving this unmeasured waveform choice. All compared holds use the same observations and the declared cold prehistory; pure delay has no backward-Euler counterpart in this study.

4.11 Prediction precision remains conditional on the mean model

Prediction bands provide a coordinate-invariant target even when the original physical coordinates have an exact null direction. Its zero-state equivalence family preserves the measured thermal response, so prediction variance can be computed on the identifiable quotient without assigning finite uncertainty to each physical parameter. By contrast, constant-input calibration with equilibrated delay states leaves a globally free delay that affects a varying future input. Those target bands are withheld. The saved finite delay-exploration envelopes are sensitivity plots without a probability interpretation.

For the jointly fitted cold representatives, mean-band widths depend strongly on the assumed covariance or residual resampling (Fig. 16). Fitted IID variance gives narrower mean bands than the displayed HAC, OU and block-residual alternatives. Wider bands under another working error law do not establish uncertainty from between-record mean discrepancy. Table 12 separates mean-band width from descriptive containment by observation bands on V01. Adding fitted future error variance does not resolve that record's transfer error. V01 is the first file in the existing sorted code map, with no selection by residual size; the complete saved summary retains every validation file, including those with better containment.

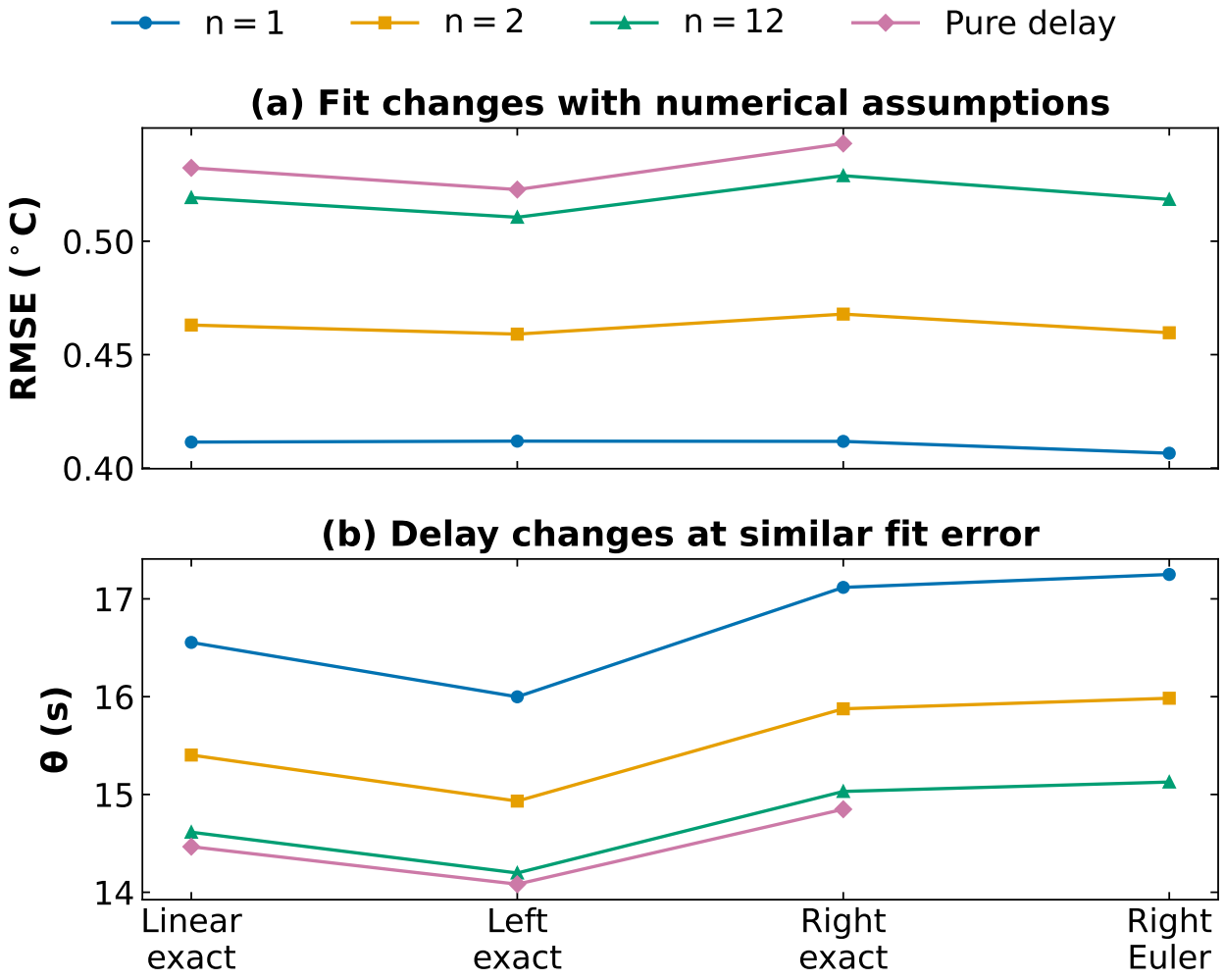


Figure 15: Different command reconstructions and propagators change the inferred delay at similar residual error. Each point is a separate joint cold fit. Linear, left and right denote the assumed hold between recorded command values. Exact uses the analytic LTI propagator for that hold; Right Euler uses the inherited backward-Euler/right-end convention at the recorded observation times. The pure-delay curve ends at the exact right-hold case. Lines join categorical assumptions and imply neither a convergence path nor uncertainty intervals.

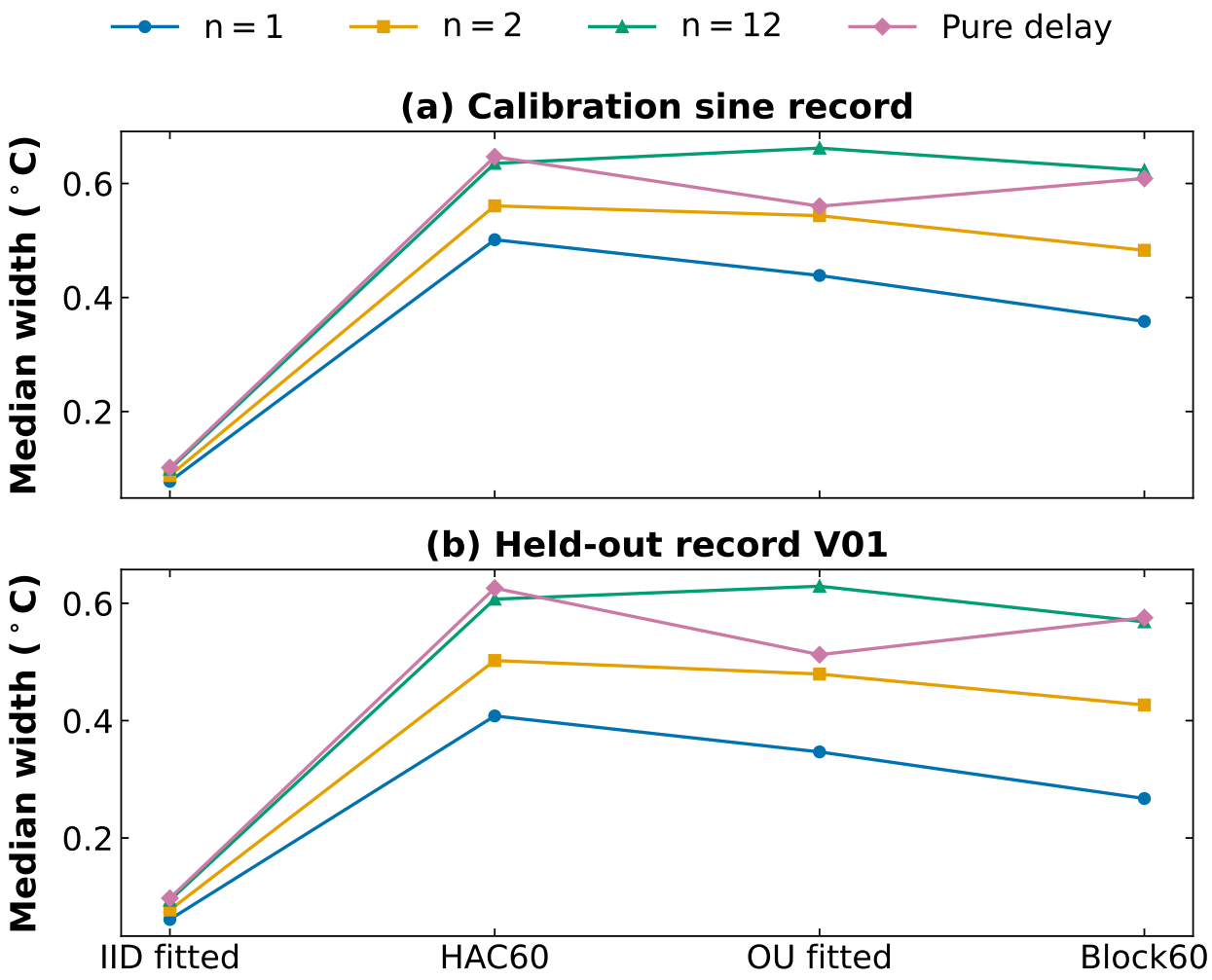


Figure 16: Conditional thermal-mean precision depends on the working error model. Points give median pointwise nominal 95% mean-band widths over the sine calibration record and held-out V01, excluding their conditioned first observations. IID fitted uses residual degrees of freedom on the identifiable quotient; HAC60 uses Bartlett bandwidth 60; OU fitted uses the observed mean/length Hessian with variance profiled; Block60 uses percentile predictions from saved joint circular-block draws. The OU means are separately fitted; other methods retain the baseline mean. Lines join methods as visual guides, without a common unconditional confidence or repeated-experiment coverage interpretation. These are mean bands, not future-observation bands.

Table 13: Selected local ensemble designs over 48 model/startup/covariance scenarios. Utility is the mean or minimum scenario-relative log improvement. Future-only ensembles use multisine as reference; ensembles with historical information use constant input, whose total information is nonsingular after adding calibration information. Criterion scales differ. Temperature is the largest independently checked value across sixteen fitted mean models through the shutdown tail. These local results do not establish global optimality, probabilistic model robustness or hardware limits.

Ensemble, information	Criterion	Δ log utility	Max. (°C)
mean, future	D	1.508	84.990
mean, history	D	1.390	84.990
mean, future	E	1.423	74.333
mean, history	E	0.754	80.792
mean, future	Delay	1.439	74.333
mean, history	Delay	0.758	80.792
mean, future	Prediction	0.861	84.990
mean, history	Prediction	0.348	84.990
min, future	D	1.294	84.990
min, history	D	1.299	84.151
min, future	E	0.560	84.990
min, history	E	0.621	75.349
min, future	Delay	0.620	73.829
min, history	Delay	0.631	75.463
min, future	Prediction	0.704	84.990
min, history	Prediction	0.300	84.990

4.12 Local input design improves declared information criteria

All parameter-design cases completed after the shutdown-grid correction, and the independent audit passed the information matrices, eigenpairs, objective values, selected-start rule and resource constraints. Each selected local solution improves its own declared reference criterion. This comparison is conditional on the fitted model points, working covariance and initialization; it does not select a physically correct model.

The ensemble designs show how the target changes the proposed experiment (Table 13 and Fig. 18). The minimum criterion targets the least favorable relative scenario; the mean criterion targets average log improvement. Historical information changes both the objective and the reference: future-only ensembles use multisine because constant input leaves delay unobservable in equilibrated scenarios, whereas adding the two historical calibration records makes the constant-reference total information nonsingular. Differences between the two panels therefore cannot be read as absolute information loss from including historical data. The determinant, minimum-eigenvalue, delay and prediction criteria also have different scales.

Figure 17 displays two selected ensemble determinant designs and their dense temperature checks. The upper temperature envelope ranges over the declared sixteen fitted mean models, without probabilities over models or parameter uncertainty. The selected inputs use the amplitude and slew limits and approach the imposed temperature constraint. The finite tail check includes the last command's ramp to zero. All selected designs pass the independent dense acceptance check, but this does not establish a continuous-time or hardware temperature bound.

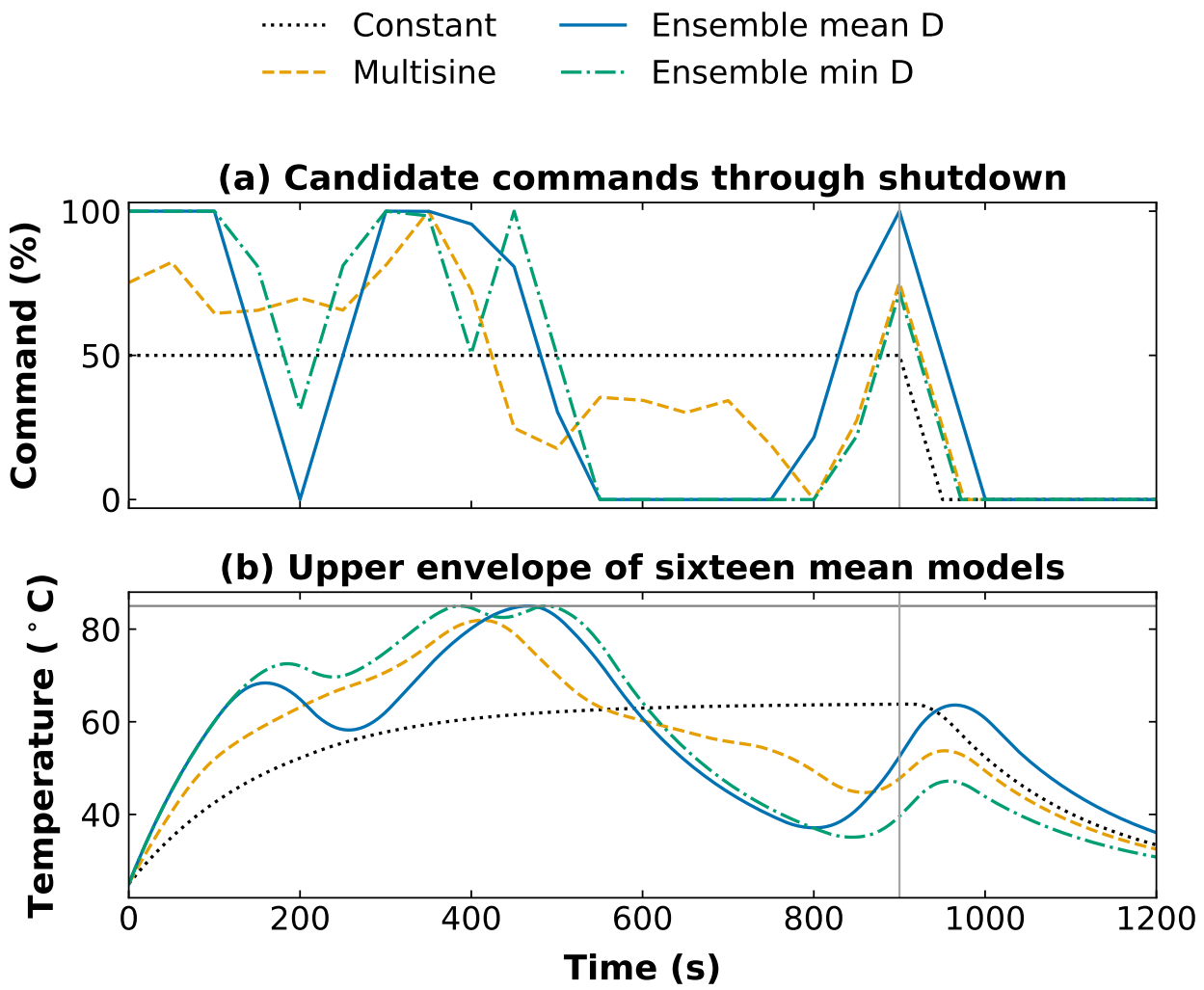


Figure 17: Information design changes the command and its predicted thermal envelope. Constant and multisine are common candidates; ensemble mean D and min D are selected determinant designs using future information only. Commands share the declared horizon, time-average dose, amplitude and slew constraints. The lower panel is the pointwise maximum temperature among sixteen fitted mean/startup combinations, not a confidence band or one physical trajectory. The vertical line marks the end of the design horizon and start of shutdown; the horizontal line is the assumed temperature acceptance limit. Calculations use a 0.25 s audit grid through the 300 s tail.

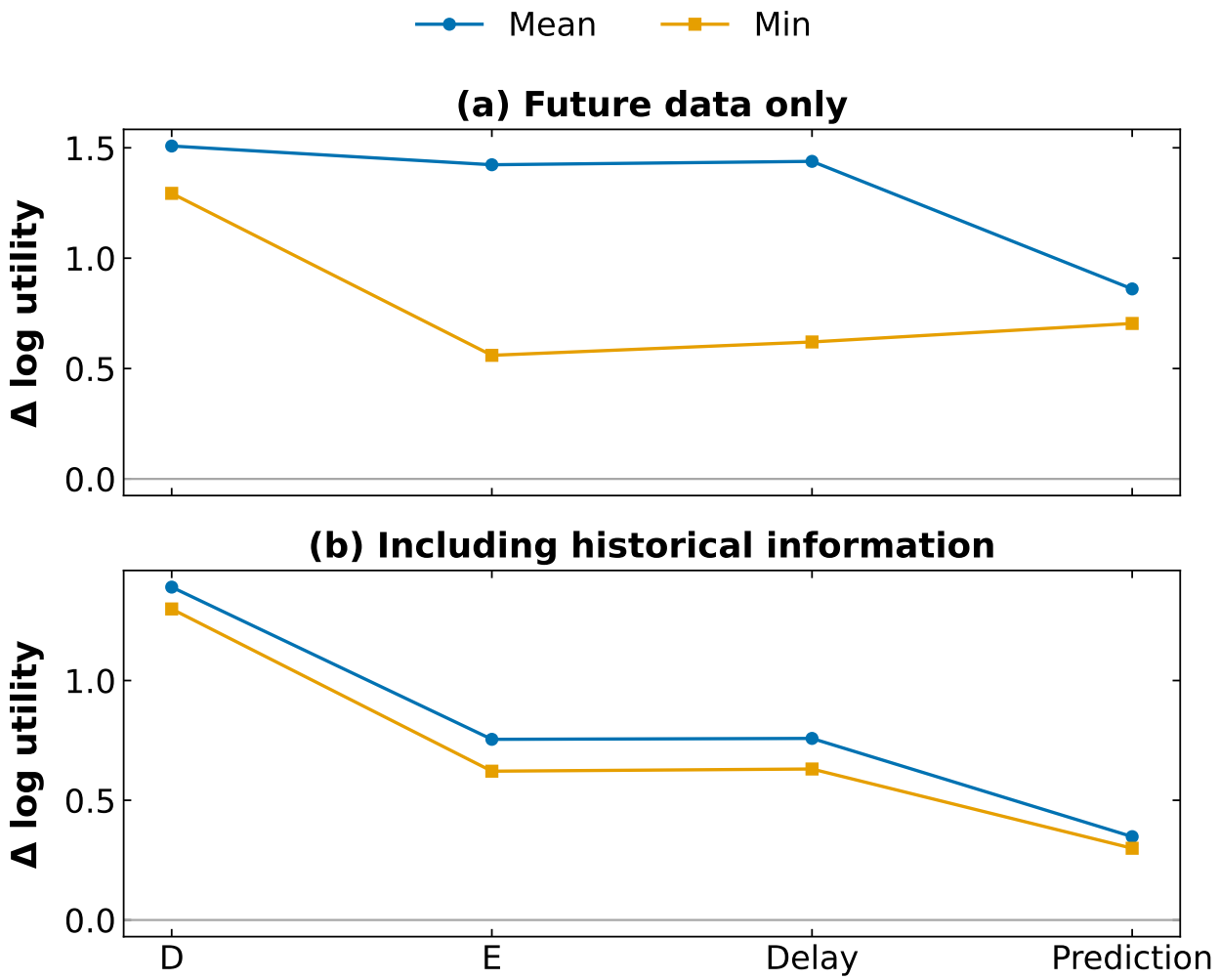


Figure 18: The design target and scenario aggregate change relative utility. Points are the independently audited selected local objective values. Mean and min aggregate scenario-relative log improvements over 48 fixed model/startup/covariance scenarios. The first panel uses multisine as reference; the second adds historical calibration information and uses constant input. Criteria have different targets and scales, so ordinate heights across criteria or reference changes do not rank absolute physical information. Lines join categorical criteria as guides. No global optimality certificate is implied.

Table 14: Fixed-mean discrimination designs compare six pairs among the four representative formulations. The objective aggregates pairwise log improvements relative to multisine, whereas the last column is the smallest raw selected pair distance; these are different comparisons. Covariances are common known working laws with marginal standard deviation 0.25 °C. Alternative-parameter refitting is evaluated separately in Fig. 19.

Startup, covariance	Aggregate	$\Delta \log D^2$	Smallest D^2
Cold, IID	mean	0.852	52.329
Cold, IID	min	0.798	52.329
Cold, OU10	mean	0.825	17.797
Cold, OU10	min	0.652	18.054
Cold, OU60	mean	0.949	59.994
Cold, OU60	min	0.628	58.797
Equilibrated, IID	mean	1.184	18.955
Equilibrated, IID	min	1.246	19.525
Equilibrated, OU10	mean	1.451	2.136
Equilibrated, OU10	min	1.324	1.564
Equilibrated, OU60	mean	2.596	6.202
Equilibrated, OU60	min	2.368	6.202

4.13 Refitting competitors changes model discrimination

The separate fixed-mean discrimination designs compare all six pairs among the representative formulations (Table 14). Their mean/min objectives concern log improvements relative to multisine for each pair. The smallest raw pair distance is a different summary and need not rank those objectives in the same order. These designs share the parameter-design resource assumptions but optimize mean separation rather than parameter information.

Allowing the competing formulation to change parameters reduces its distance from the fixed one-stage generating mean (Fig. 19). Under equilibrated constant input, the closest-found distance vanishes to numerical precision for every alternative and working covariance. The matched gain and time constant give an analytic zero-distance witness because the delay factor is absent. Positive separation at the original fitted alternative parameters had therefore overstated the information in that input about model order.

The selected varying inputs retain larger closest-found distances than multisine in this bounded directional comparison, including after refitting. That result concerns a fixed one-stage generating mean against three alternatives, rather than every possible generating formulation or parameter point. Each smallest converged distance is an upper bound on the global closest alternative. The ideal equal-prior Gaussian error formula for two fixed means cannot be reused as a power calculation after fitting those alternatives or allowing the observed between-record discrepancy.

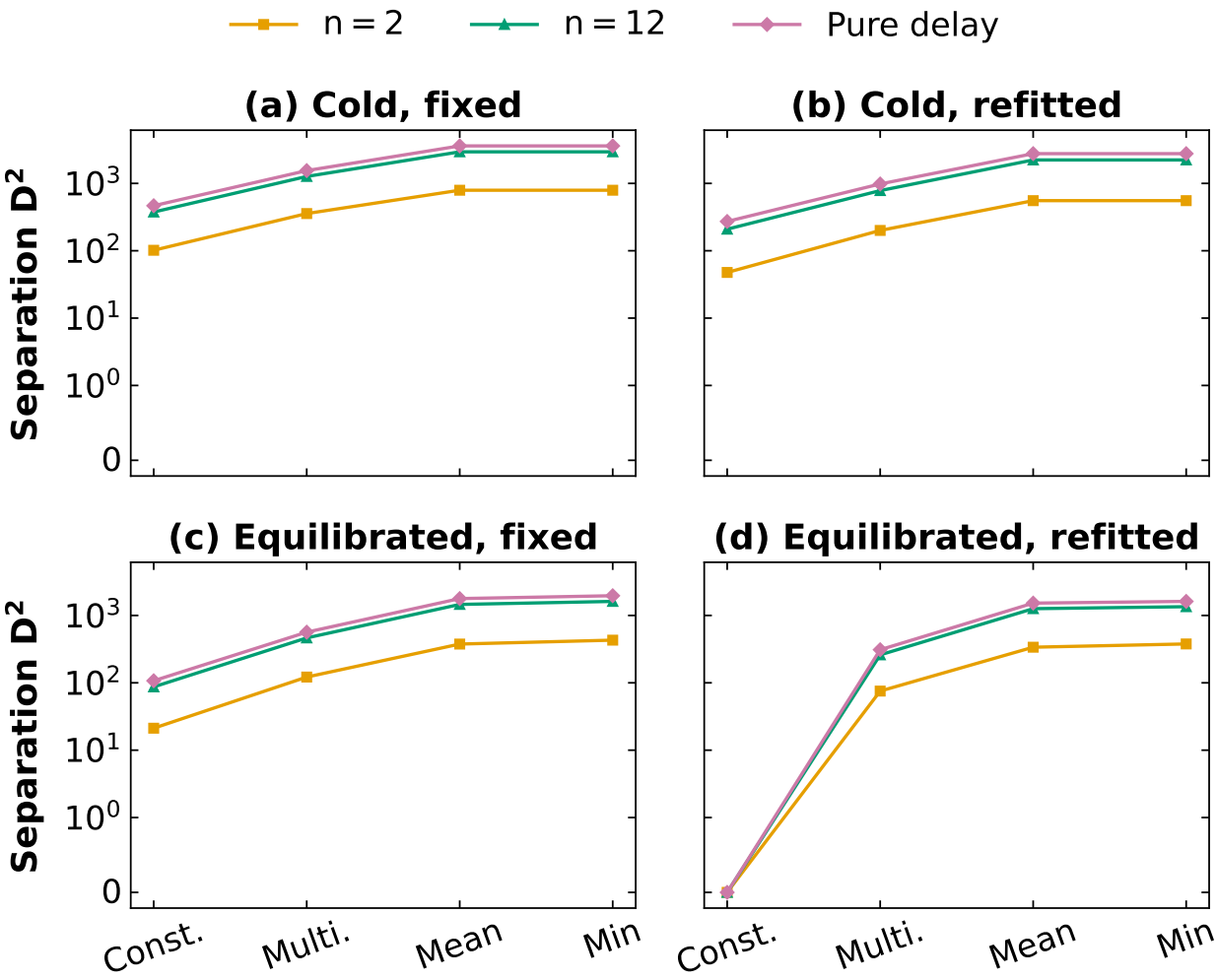


Figure 19: Parameter refitting can remove apparent model-order separation. The fixed one-stage generating mean is compared with the three indicated alternative formulations under the first declared covariance, IID Gaussian error with marginal standard deviation 0.25 °C. Rows use cold and input-equilibrated delay states. Left panels retain each alternative's original fitted parameters; right panels use its smallest converged 24-start distance. Const. and Multi. are common candidates; Mean and Min are the selected fixed-mean discrimination inputs. Symmetric-log axes are linear below one, retaining vanishing distances without a positive plotting floor. Lines join categorical waveforms. The complete saved comparison also includes both correlated working covariances. Distances after refitting are upper bounds on global minima, without a real-data testing-power interpretation.

5 Discussion

The transfer-function analysis places a limit on what a better input can achieve. Under the single measured output and fixed initial conditions, excitation cannot separate the original model's equivalent physical parameter combinations. Estimating the three transfer coefficients or measuring an additional independent quantity is a different scientific task from narrowing a four-parameter local covariance. The existing data include a second sensor, but its usefulness for breaking this equivalence requires a model that connects that sensor to the states; the second trace alone does not establish that connection.

The reduced conductance–capacity parameters require an effective-model interpretation. The gain fixes $U_{\mathrm{a}} = c/K$ only conditional on the adopted heater conversion. The reduced heat capacity $C_{\mathrm{p}} = c\tau/K$ is tied to the selected thermal pole; it is not automatically the heater heat capacity $C_{\mathrm{p,H}}$ of the original model. The original model's equivalence class permits $C_{\mathrm{p,H}}$ to vary while both measured transfer coefficients and temperature predictions are fixed.

Initialization is part of the model specification. A delay that appears estimable under cold delay states may become unobservable under equilibrated delay states and constant input. Future laboratory records should include the heater history before the fitting window and an independent ambient measurement. These records would reduce dependence on assumptions made at the first sample.

The conditional delay and physical-family-boundary intervals vary with residual block length (Figs. 9 and 7). The circular bootstrap resamples centered residuals within each calibration record and retains dependence within each block. The Gaussian comparator uses a fixed hypothetical noise scale, whereas the block methods use empirical centered residuals. Their contrast can therefore reflect both error scale and serial dependence. Centering also removes each calibration record's mean residual. This construction therefore does not estimate the between-record offsets seen in validation, even when its percentile intervals are wider than those from independent Gaussian simulation. Selecting the widest of these conditional intervals would not supply a probability model for the omitted discrepancy.

Fitting an OU correlation length changes the weighting of thermal trajectories and their local uncertainty, but the remaining negative innovation correlation challenges that combined mean/covariance model. The fitted length cannot yet be interpreted as a measured physical disturbance time scale. Neither a Gaussian likelihood improvement nor a HAC variance calculation establishes that the observed error process satisfies the assumptions needed for confidence coverage.

Transfer to the held-out files is a stronger concern than the small differences among the representative cold model orders. The pooled error and its file-mean decomposition (Table 3) show why narrower calibration intervals alone would be insufficient evidence for using a formulation on a new record. Subtracting each held-out mean error would use information unavailable before that record was observed. Independent ambient, command-history and heater-power measurements could help distinguish candidate explanations; the supplied filenames and temperature offsets do not identify one.

The controlled coverage experiment isolates one consequence of a wrong error law: IID likelihood intervals can fail even when the thermal mean is exactly correct. HAC improves the comparison without reaching the nominal reference at the longer tested correlation length. The finite-grid profile diagnostic also prevents treating every interpolated threshold set as an exact numerical confidence set. Neither result supplies coverage for the real-data bootstrap, whose centering removes the between-record mean errors seen in validation.

The ambient/history and command-hold studies change a different part of the inference. Ambient offsets lower calibration error but reach imposed bounds; command reconstruction shifts the selected

delay with less visible change in RMSE. These sensitivities can exceed the apparent precision of a chosen conditional error model. Recording independent ambient and heater history would therefore answer a question that collecting more samples under the same unverified initialization cannot settle.

The prediction calculation preserves the distinction between a redundant physical representation and an unresolved future response. Quotient prediction variance remains finite along the original physical equivalence family, whereas an unobserved delay from a constant-input calibration can change a later varying response. A pseudoinverse alone cannot decide which of those situations applies. The prospective calculations specify what a proposed experiment is meant to learn. Parameter-information and model-separation criteria choose different inputs; initialization and the working covariance remain part of each design. Refitting the alternatives removes constant-input separation in the equilibrated case, while the selected varying inputs retain separation in the bounded one-stage-generating comparison. None of those designs can remove the original physical equivalence family under the same modeled output.

The next laboratory study should record an independent ambient trace, the pre-experiment command history and heater-power calibration, then test a selected varying input against held-out predictions declared beforehand. A second sensor could address the physical ambiguity only after its observation equation is specified and checked. Those measurements would test explanations for the transfer errors that the current conditional intervals and fixed-point input designs leave unresolved.

6 Data, code, and research status

The repository source is `https://github.com/dowlinglab/pyomo-doe`. The study is under `skunk_works/delay_study` on branch `reformulation-exploration`. Phase manifests identify immutable source commits, data hashes, configuration, environment, seeds, and numerical outcomes. Release receipts identify the source revision, compiled report checksum, numerical phase manifests and artifact checks. The supplied raw data and inherited source were preserved.

An OpenAI Codex agent performed the notebook exports, implemented the numerical analysis, ran the computations, searched and inspected literature, and drafted this document. This disclosure describes scientific and software work as well as writing assistance. The document has not received human scientific review. No laboratory hardware experiment was performed for this study, and no target journal or submission policy has been selected.

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A Structural derivations

A.1 Delay-chain moments and pure-delay limit

Let $\lambda_n = n/\theta$ be the delay-state rate. The normalized chain kernel for $t \geq 0$ is

$$h_n(t) = \frac{\lambda_n^n t^{n-1} e^{-\lambda_n t}}{(n-1)!}.$$

Its Laplace transform is $(1 + s/\lambda_n)^{-n}$. Differentiating that transform at zero gives mean $n/\lambda_n = \theta$ and variance $n/\lambda_n^2 = \theta^2/n$. Consequently, the normalized complete thermal response has impulse-time mean $\tau + \theta$ and variance $\tau^2 + \theta^2/n$. These are properties of the model kernel; they are not estimates of a measured residence-time distribution. For fixed s in the domain of convergence,

$$\log(1 + \theta s/n)^{-n} = -n \log(1 + \theta s/n) \longrightarrow -\theta s.$$

This calculation supplies the pure-delay limit of Eq. (4).

A.2 An explicit physical-parameter equivalence class

Define the two positive pole time constants $t_\mathrm{f} < t_\mathrm{s}$ by $a = t_\mathrm{f} + t_\mathrm{s}$ and $b = t_\mathrm{f} t_\mathrm{s}$. For fixed positive transfer coefficients (K, a, b), choose a family coordinate $r = C_{\mathrm{p,S}}/U_\mathrm{b}$ between these pole time constants. The physical parameters

$$U_\mathrm{a} = \frac{c}{K}, \qquad C_{\mathrm{p,H}} = \frac{bU_\mathrm{a}}{r}, \qquad C_{\mathrm{p,S}} = U_\mathrm{a}\left(a - r - \frac{b}{r}\right), \qquad U_\mathrm{b} = \frac{C_{\mathrm{p,S}}}{r} \tag{9}$$

produce exactly the same transfer function for every $r \in (t_\mathrm{f}, t_\mathrm{s})$. Positivity follows from $a - r - b/r = -(r - t_\mathrm{f})(r - t_\mathrm{s})/r > 0$. Substitution into Eq. (8) recovers K, a, and b exactly. The conductance U_a is identifiable through K; the other three parameters vary along this equivalence class. No input waveform can distinguish members of this class through this output under the stated initial conditions.

Positivity constrains this continuum to finite ranges in some coordinates. For example, its heat capacities obey

$$U_\mathrm{a} t_\mathrm{f} < C_{\mathrm{p,H}} < U_\mathrm{a} t_\mathrm{s}, \qquad 0 < C_{\mathrm{p,S}} \leq U_\mathrm{a}\left(a - 2\sqrt{b}\right).$$

The second upper bound occurs at $r = \sqrt{b}$. The coupling conductance satisfies

$$0 < U_{\mathrm{b}} \le U_{\mathrm{a}} \left(\frac{a^2}{4b} - 1 \right),$$

with its maximum at $r = 2b/a$. Thus bounded physical ranges or bounded exploration grids do not establish structural identifiability. Structural uniqueness must be checked separately from the numerical width of a confidence set.

The one-stage reduced model has $a = \tau + \theta$ and $b = \tau\theta$. For a fixed transfer function, its two pole labels therefore assign reduced capacities $C_{\mathrm{p}} = U_{\mathrm{a}}\tau$ equal to $U_{\mathrm{a}}t_{\mathrm{f}}$ and $U_{\mathrm{a}}t_{\mathrm{s}}$. These values are precisely the infimum and supremum of the original heater capacity, rather than interior members of its strictly positive family. A capacity obtained from either reduced pole label cannot be identified with a unique original $C_{\mathrm{p,H}}$. The boundary functions themselves are identifiable from (K, a, b) and their sampling uncertainty can be propagated without selecting a family coordinate r.

The strictly positive original parameters imply distinct poles. To see this, write $h = C_{\mathrm{p,H}}/U_{\mathrm{a}}$, $r = C_{\mathrm{p,S}}/U_{\mathrm{b}}$, and $q = C_{\mathrm{p,S}}/U_{\mathrm{a}}$. Then $a = h + r + q$ and $b = hr$, giving $a > h + r \ge 2\sqrt{b}$. Coincident poles are attained only as a limiting case outside this strictly positive four-parameter family.

A.3 Constant input with equilibrated delay states

If $u(t) = u_0$ and $z_i(0) = u_0$, every delay-state derivative in Eq. (1) vanishes. The predicted temperature then depends on K and τ alone, and its derivative with respect to θ is zero at every observation time. A constant prehistory gives the same conclusion for Eq. (5). This experiment cannot estimate the delay even when the thermal transient is well measured. Cold delay states under the same post-start input retain a transient that depends on θ.

A.4 Coordinate changes and information

For positive parameters, write $\boldsymbol{\eta} = (\log K, \log\tau, \log\theta)^{\top}$ and $\boldsymbol{\xi} = (\log U_{\mathrm{a}}, \log C_{\mathrm{p}}, \log\theta)^{\top}$. The Jacobian of the map from $\boldsymbol{\eta}$ to $\boldsymbol{\xi}$ is the constant matrix

$$M = \begin{pmatrix} -1 & 0 & 0 \\ -1 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \qquad M^{-1} = M.$$

For a nonsingular covariance C_η, the transformed local covariance is $C_\xi = MC_\eta M^{\top}$. Sensitivity rank and prediction-space uncertainty are invariant under this invertible transformation. Numerical eigenvalues, eigenvectors, and condition numbers generally change with the coordinate metric. The matrix satisfies $|\det M| = 1$, so log-coordinate information determinants at the same model point are equal in this particular transformation. This equality applies to future-input information and to any historical information transformed by the same congruence. It therefore preserves the D-design objective and its input gradient. The delay target is unchanged because the third coordinate is unchanged, and a prediction loss is preserved when its target matrix transforms with the sensitivities. The minimum information eigenvalue does not have this invariance.

A.5 Repeated poles and a singular local information matrix

At a repeated thermal and delay pole, $\tau = \theta/n$, the transfer function is $(1 + \tau s)^{-(n+1)}K$. The derivatives in unscaled coordinates satisfy

$$\frac{\partial \log G_n(s)}{\partial \tau} = -\frac{s}{1 + \tau s}, \qquad \frac{\partial \log G_n(s)}{\partial \theta} = -\frac{s}{1 + \theta s/n}.$$

These two derivatives coincide at the repeated pole. The corresponding log-coordinate sensitivity columns are proportional with ratio n, producing a singular local information matrix under zero-state excitation. For $n > 1$, the rational transfer function still uniquely specifies the distinct single and repeated poles away from their coincidence; at coincidence there is one common pole value. Local derivative rank loss at this exceptional point therefore does not by itself establish a continuum of exactly equivalent parameters. Nonregular local inference and structural nonuniqueness are separate questions. The exact matrix-exponential simulator avoids a partial-fraction formula that would divide by the pole separation.

The exceptional point also changes the order of local objective curvature. Let $t_c > 0$ be the common pole time and perturb $\tau = t_c + \delta$ and $\theta = nt_c - \delta$, holding the total mean time fixed. For small δ,

$$\log \frac{G_n(s;\delta)}{G_n(s;0)} = \frac{n+1}{2n}\,\delta^2 \left(\frac{s}{1+t_c s}\right)^2 + O(\delta^3).$$

The linear term cancels. The temperature difference is therefore second order in this direction and its squared prediction distance is fourth order, subject to a nonzero response coefficient for the chosen input. For $n = 1$, exchanging δ and $-\delta$ exchanges pole labels exactly. For $n > 1$, the repeated factor's multiplicity generally distinguishes those two perturbed models.

A.6 Moment coordinates

Define the total mean response time $m = \tau + \theta$ and the delay fraction $f = \theta/m$. The inverse map is $\tau = m(1-f)$ and $\theta = mf$ with $m > 0$ and $0 < f < 1$. If $\ell = \log[f/(1-f)]$, the Jacobian from $(\log m, \ell)$ to $(\log \tau, \log \theta)$ is

$$\begin{pmatrix} 1 & -f \\ 1 & 1-f \end{pmatrix}.$$

This determinant-one map provides an alternative optimization metric without changing model predictions. Its rectangular bounds generally differ from the original rectangular bounds in τ and θ.

The impulse-time variance $v = \tau^2 + \theta^2/n$ provides another summary of the transfer shape. Given m and v, the possible delay values solve

$$\left(1 + \frac{1}{n}\right)\theta^2 - 2m\theta + (m^2 - v) = 0.$$

Two positive solutions can therefore share the first two moments. For $n = 1$ they exchange the two poles and give the same full transfer function. For $n > 1$, higher moments can distinguish the branches. The minimum feasible variance, $v = m^2/(n+1)$, occurs at the repeated-pole condition above. These identities motivate comparing coordinate conditioning and response moments, while retaining the full dynamic model for estimation.

A.7 A measurement that could break the original equivalence

Suppose an additional sensor measured the model's heater state T_H directly. Let $Y_H(s)$ and $Y_S(s)$ denote the Laplace transforms of the heater and sensor temperature deviations from ambient. For zero initial deviations, the second balance in Eq. (7) gives

$$\frac{Y_H(s)}{Y_S(s)} = 1 + \frac{C_{p,S}}{U_b}s = 1 + rs,$$

for nonzero transfer responses. Identifying this additional time constant r would select one member of Eq. (9) and hence recover all four physical parameters. The supplied second TC-Lab temperature trace has not been established as a direct measurement of this model's heater state.

A.8 Dependent-residual sandwich calculation

For a full-rank sensitivity matrix, let $B = (J^\top J)^{-1}$ and define the score contribution $\boldsymbol{q}_t = J_t^\top e_t$ at residual index t within an experiment. With lag bandwidth L, the matrix used between the two factors of B is

$$S = \sum_t \boldsymbol{q}_t \boldsymbol{q}_t^\top + \sum_{j=1}^{L} \left(1 - \frac{j}{L+1}\right) \sum_{t=j+1}^{N} \left(\boldsymbol{q}_t \boldsymbol{q}_{t-j}^\top + \boldsymbol{q}_{t-j} \boldsymbol{q}_t^\top\right).$$

For joint calibration we formed each lagged sum within its experiment, then added the experiment contributions; no lag joined separate experiments. The reported covariance is $[N/(N-k)]BSB$, where k is the parameter count and N is the total conditional residual count. The scalar degrees-of-freedom adjustment is an implementation choice. Bartlett weighting follows the positive-semidefinite construction of Newey and West (1987); it does not establish that the TC-Lab score process satisfies the paper's consistency assumptions. Lags here count retained observations rather than elapsed seconds.

B Calibration, initial-state and residual assumptions

B.1 Heater calibration introduces another exact ambiguity

The conversion $c = \alpha P$ is fixed in the baseline study. If it is unknown, the transformation

$$(c, U_\mathrm{a}, C_\mathrm{p}) \longmapsto (\lambda c, \lambda U_\mathrm{a}, \lambda C_\mathrm{p}), \qquad \lambda > 0,$$

preserves $K = c/U_\mathrm{a}$, $\tau = C_\mathrm{p}/U_\mathrm{a}$, and every reduced-model prediction under the same initial conditions. In the original model, scaling $c, U_\mathrm{a}, U_\mathrm{b}, C_\mathrm{p,H}$ and $C_\mathrm{p,S}$ together has the same effect. Input design cannot identify this absolute thermal scale from temperature and percentage command alone.

Let C_io be a covariance matrix for $(\log K, \log \tau, \log \theta)$ and suppose an independent heater calibration has log-variance v_c. With the transformation matrix M defined in Appendix A, the corresponding physical log-covariance is

$$C_\mathrm{phys} = M C_\mathrm{io} M^\top + v_c \boldsymbol{v}\boldsymbol{v}^\top, \qquad \boldsymbol{v} = (1, 1, 0)^\top.$$

This covariance identity is exact in logarithmic coordinates under the stated independence assumption; Gaussianity is not needed for the covariance itself. It adds correlated uncertainty to conductance and capacity while leaving delay unchanged. A hypothetical lognormal calibration with coefficient of variation r_c has $v_c = \log(1 + r_c^2)$. Such a sensitivity calculation must be labelled as assumed calibration uncertainty: these data do not supply r_c.

B.2 Ambient and history are nuisance quantities

For either prescribed delay-state initialization, hold the observed initial sensor temperature T_0 fixed and vary ambient temperature. Linearity gives

$$\frac{\partial T(t)}{\partial T_\mathrm{amb}} = 1 - e^{-t/\tau}.$$

An ambient error therefore persists rather than averaging away with additional samples in one record. Under constant input u_0 and equilibrated delay states,

$$T(t) = T_0 e^{-t/\tau} + (T_\mathrm{amb} + K u_0)(1 - e^{-t/\tau}).$$

Only the combination $T_{\mathrm{amb}} + Ku_0$ is identified from this response if ambient is free, and delay is absent. Independent ambient information and a varying input address different ambiguities.

Another restricted history family sets every chain state initially to γu_0, where $0 \le \gamma \le 1$. For fixed parameters, its response is exactly $(1-\gamma)T_{\mathrm{cold}} + \gamma T_{\mathrm{equilibrated}}$. The same interpolation describes a constant pre-window input between zero and u_0 in the pure-delay model. This family does not represent every possible heater history. Its derivative with respect to γ is the difference of the two endpoint trajectories. If these nuisance sensitivities resemble the delay sensitivity, fixing history can make delay appear more precise than a fit that allows history to vary.

More generally, let $A = W^{1/2}J_p$ and $B = W^{1/2}J_a$ be the mean sensitivities for parameters of interest and nuisance parameters, respectively, under a fixed positive-definite observation weight W. Profiling unrestricted local nuisance directions leaves information $A^\top(I - BB^+)A$, where B^+ is the Moore–Penrose pseudoinverse. This projection cannot increase information. Bounds or independent nuisance measurements alter the problem and must be stated.

B.3 Free history can preserve the exchanged pole labels

For the one-stage model, write $y = T - T_{\mathrm{amb}}$, $A_0 = T_0 - T_{\mathrm{amb}}$, and $z(0) = \gamma u_0$, where $u_0 = u(0)$. Taking the one-sided Laplace transform of the two state equations gives

$$(1 + \tau s)(1 + \theta s)Y(s) = KU(s) + K\theta\gamma u_0 + \tau A_0(1 + \theta s).$$

Exchange τ and θ while keeping K, ambient and the observed initial temperature fixed. The denominator and the coefficient of s in the initial-state term stay the same. Equality of the remaining constant terms requires, for $u_0 \neq 0$,

$$\gamma' = \frac{\theta\gamma + A_0(\tau - \theta)/(Ku_0)}{\tau}. \tag{10}$$

This transformation preserves the entire trajectory for any subsequent input. It is admissible only if $\gamma' \in [0, 1]$ and both parameter vectors satisfy their fit bounds. For a joint fit, each record needs its own admissible transformed history fraction. When ambient has an offset, A_0 uses that record's actual fitted ambient.

At $A_0 = 0$, equation (10) reduces to $\gamma' = \gamma\theta/\tau$. A prescribed equilibrated history can separate pole labels that become indistinguishable when history is free. If $u_0 = 0$, history fractions have no effect in this restricted family; the exchange requires $A_0(\tau - \theta) = 0$ instead. A local full-rank sensitivity matrix cannot exclude an admissible disconnected pole-label solution.

B.4 Constant input admits a gain–ambient–history continuum

For a constant command u_0, define the delay-state deficit

$$d_n(t) = e^{-nt/\theta} \sum_{k=0}^{n-1} \frac{(nt/\theta)^k}{k!}$$

for an n-stage chain, and $d_\infty(t) = \mathbf{1}_{\{t<\theta\}}$ for pure delay. With the uniform initial history fraction γ, the thermal forcing is $Ku_0[1 - (1-\gamma)d_n(t)]$. Thus

$$T(t) = T_0 e^{-t/\tau} + C(1 - e^{-t/\tau}) - \frac{Bu_0}{\tau} \int_0^t e^{-(t-v)/\tau} d_n(v)\, dv, \qquad C = T_{\mathrm{amb}} + Ku_0, \quad B = K(1-\gamma).$$

The same expression applies to pure delay with d_∞. Holding τ, θ, B and C fixed gives the exact family

$$K' > 0, \qquad \gamma' = 1 - \frac{K(1-\gamma)}{K'}, \qquad T'_{\text{amb}} = T_{\text{amb}} + u_0(K - K'). \tag{11}$$

Every admissible member predicts the same measured-temperature trajectory. The gain box, ambient-offset bounds and $0 \le \gamma' \le 1$ restrict the family. At an interior point they still permit a continuum. With $\gamma = 1$, the delay is also absent from the response. Varying the input generally breaks equation (11), since the time-varying gain forcing cannot be absorbed into a constant ambient offset. Independent state-propagation checks verify this family for one-, two- and twelve-stage chains and pure delay, and the pole exchange for a varying input.

B.5 Serial dependence alone does not prove least-squares bias

For fixed input and known initial conditions, suppose observations have the correct mean $\boldsymbol{m}(\boldsymbol{p}_0)$ and additive error $\boldsymbol{\epsilon}$ with $E\boldsymbol{\epsilon} = 0$ and covariance Σ. For any candidate parameter vector $\boldsymbol{p}$,

$$E\|\boldsymbol{m}(\boldsymbol{p}) - \boldsymbol{m}(\boldsymbol{p}_0) - \boldsymbol{\epsilon}\|^2 = \|\boldsymbol{m}(\boldsymbol{p}) - \boldsymbol{m}(\boldsymbol{p}_0)\|^2 + \operatorname{tr}\Sigma.$$

Correlated errors change sampling uncertainty and efficiency without moving the minimum of this expected output-error objective. This identity does not imply an unbiased finite-sample nonlinear estimator. It also does not apply unchanged when noisy observations enter regressors, the initial observation creates a nonzero conditional error mean, or the thermal mean is misspecified. The observed TC-Lab residual offsets and held-out errors require those possibilities to remain open.

B.6 An OU filter can anticorrelate an IID component

Consider a fixed OU correlation factor $0 < \phi < 1$ at regular sampling and an IID component v_i with variance s_v^2. Away from the first retained innovation, the unnormalized filtered component is $w_i = v_i - \phi v_{i-1}$. Independence gives

$$\operatorname{Var}(w_i) = (1+\phi^2)s_v^2, \qquad \operatorname{Cov}(w_i, w_{i-1}) = -\phi s_v^2, \qquad \rho_w(1) = -\frac{\phi}{1+\phi^2}.$$

Dividing by the constant OU innovation scale does not change this correlation. It approaches $-1/2$ as $\phi \to 1$. Thus an IID component inside a residual process can produce negative filtered correlation even when slower components are present. This calculation concerns one component and a fixed regular-sample filter. It neither establishes that the TC-Lab has this decomposition nor assigns the observed correlations to sensor noise rather than mean error.

C Chain convergence at the causal onset

For a cold constant step of magnitude u_0 and zero initial thermal deviation, let D_n have an Erlang distribution with mean θ and variance θ^2/n. The temperature rise relative to ambient is

$$y_n(t) = Ku_0 E\left[\left(1 - e^{-(t-D_n)/\tau}\right)\mathbf{1}_{\{D_n \le t\}}\right].$$

At $t = \theta$, the pure-delay response is zero while every finite chain has a positive response. The central limit theorem and a first-order expansion of the exponential give

$$\sqrt{n}\, y_n(\theta) \longrightarrow \frac{Ku_0\theta}{\tau\sqrt{2\pi}}.$$

To justify the expectation limit, the standardized Erlang variables have unit second moments, so their positive parts are uniformly integrable. The exponential remainder is bounded by a constant times $E(\theta - D_n)^2/\tau^2 = O(n^{-1})$. This is an onset error of order $n^{-1/2}$, even when smooth parts of the response converge faster. A small average trajectory error can therefore hide a larger difference near the delay's causal onset. The saved onset calculation checks the coefficient using an independently tested Erlang-CDF expression.

D Frequency sensitivity and local excitation requirements

The following identities are derived for the transfer functions in this study. They explain an ideal periodic steady-state experiment. They do not replace the finite-horizon, initialized simulations or their working covariance models. For positive frequency ω, let $G = G_n(i\omega)$ and define the complex log-transfer sensitivity vector $\boldsymbol{h} = G^{-1}\partial G/\partial \boldsymbol{\eta}$, where $\boldsymbol{\eta} = (\log K, \log \tau, \log \theta)$. Direct differentiation gives

$$\boldsymbol{h}_n(\omega) = \left(1,\ \frac{-i\omega\tau}{1+i\omega\tau},\ \frac{-i\omega\theta}{1+i\omega\theta/n}\right)^{\top}. \tag{12}$$

For the pure-delay model, the third component is $-i\omega\theta$. At zero frequency, only the gain sensitivity remains. At low frequency,

$$\log G_n(i\omega) = \log K - i\omega(\tau+\theta) - \tfrac{1}{2}\omega^2(\tau^2 + \theta^2/n) + O(\omega^3).$$

Thus the leading phase change depends on the sum of the thermal and delay lags. Their separation depends on higher-order frequency behavior; a small low-frequency fit error can coexist with poor precision for each lag. The θ^2/n term is absent in the pure-delay expansion.

To separate this local frequency geometry from startup information, consider a single periodic input $u(t) = u_0 + A\cos\omega t$ after transients have decayed. Let $\boldsymbol{d} = \partial G(i\omega)/\partial\boldsymbol{\eta}$ and suppose the N observed phases have zero sine/cosine sums, zero sine–cosine cross sum, and equal sine/cosine sums of squares $N/2$. With gain direction $\boldsymbol{e}_1 = (1,0,0)^{\top}$ and independent errors of known variance σ^2, the exact phase-sum information is

$$F = \frac{N}{\sigma^2}\left[(u_0K)^2\boldsymbol{e}_1\boldsymbol{e}_1^{\top} + \frac{A^2}{2}\operatorname{Re}(\boldsymbol{d}\boldsymbol{d}^{*})\right], \tag{13}$$

where $*$ denotes conjugate transpose. A single tone without its constant (DC) offset contributes a matrix of rank at most two for the three real parameters. With nonzero offset and amplitude, the DC term supplies a separate gain direction: its constant sensitivity is orthogonal to the oscillatory components over these phases. The determinant that tests independence of the lag components is

$$\begin{vmatrix} \operatorname{Re} h_\tau & \operatorname{Re} h_\theta \\ \operatorname{Im} h_\tau & \operatorname{Im} h_\theta \end{vmatrix} = \frac{\omega^3\tau\theta(\tau - \theta/n)}{(1+\omega^2\tau^2)(1+\omega^2\theta^2/n^2)}. \tag{14}$$

It is nonzero away from $\tau = \theta/n$. In this ideal experiment, a DC component and one nonzero tone therefore give local full rank at a regular parameter point. For pure delay the determinant is $\omega^3\tau^2\theta/(1+\omega^2\tau^2) > 0$. At the finite-chain repeated pole, $h_\theta = nh_\tau$ at every frequency; adding frequencies cannot remove that first-order rank deficiency. These are local differential statements. They do not remove the cold one-stage global pole-label ambiguity, calibrate a confidence interval, or establish an optimal input under amplitude, sampling and temperature constraints.

The factor G in $\boldsymbol{d} = G\boldsymbol{h}$ also matters: information depends on output sensitivity, rather than phase sensitivity alone. The time-domain designs evaluate that output sensitivity with retained

observations, startup assumptions and a specified covariance. Constant-command startup records can carry transient information that is deliberately absent from the periodic steady-state calculation. Analytic derivatives, the determinant and the phase-sum information were checked against parameter perturbations, numerical determinants and explicit real sensitivity matrices. These checks contain no additional TC-Lab fits.

E Comparison of theoretical assumptions

The review covers inspected passages from seven cited works. Two library carriers supply the profile paper and a continuous-time identification chapter; five further primary PDFs were inspected outside the ingested library. This counts works and carriers, without implying that every page was read. The source and search records retain exact queries, dates, versions, page locations and access gaps. Table 15 separates the inspected theory from its use in this study.

Table 15: Questions, source assumptions and the scope of transfer to TC-Lab. Page numbers are physical PDF pages in the inspected source. The last column gives this study's interpretation rather than a result established by the cited work.

Question and source	Inspected theory	Conditions or distinction	TC-Lab use and limit
What produces a residual? Garnier et al. (2008), pp. 27–29	The input-driven response and residual process are separate descriptions. Residuals can combine measurement error, disturbances and model inadequacy.	The chapter provides an identification framework, without selecting a correct residual mechanism for these experiments.	Treat covariance fits as working assumptions; examine transfer to other records separately.
Can parameters be separated? Raue et al. (2009), pp. 1–5	Profile likelihood fixes a parameter and reoptimizes nuisance parameters; its shape can differ from local quadratic curvature.	A likelihood and a numerical exploration are required. A sampled crossing alone cannot certify every nuisance minimum or mode.	Combine profiles with exact transfer equivalence and sensitivity rank; retain disconnected components.
What should information target? Hjalmarsson (2009), pp. 8–9	Application loss and confidence regions can concern useful parameter combinations rather than uniform precision in every coordinate.	The inspected prediction-error information formula assumes the true system is in the model set and the predictor is differentiable.	Target delay or thermal prediction while checking mean-model adequacy and the chosen parameter metric.
How is a bounded design optimized? Manchester (2010), pp. 1–5	Finite-horizon amplitude-constrained information design is nonconvex; the source develops a convex relaxation.	The inspected source is the eight-page author preprint. Its relaxation guarantees concern its own construction and assumptions.	Our multistart nonlinear solutions have no inherited approximation or global-optimality guarantee.

Question and source	Inspected theory	Conditions or distinction	TC-Lab use and limit
What does a chain represent? Hurtado and Kirosingh (2019), pp. 11–12	Erlang dwell times admit a first-order differential-equation chain, with equivalence to an integral description.	The source treats mean-field transitions. Our constant-rate input kernel uses the corresponding Erlang construction.	Derive kernel moments and limits; a fitted stage count alone does not establish a thermal transport mechanism.
How can serial dependence affect covariance? Newey and West (1987), pp. 1–3	Bartlett-weighted long-run covariance estimation is positive semidefinite; consistency has explicit score, moment and dependence conditions.	Asymptotic bandwidth conditions are distinct from choosing several fixed bandwidths for a finite dataset.	Compare HAC sensitivity without treating it as proof of coverage or a correction for mean-model bias.
What dependence does resampling retain? Politis and Romano (1994), pp. 1–3	Periodic extension supports circular resampling. Remark 2 describes the fixed-length relative of the geometric stationary-bootstrap scheme.	The schemes differ. Equal marginal weights at record edges do not establish nonlinear interval coverage with model discrepancy.	Vary fixed block lengths and retain within-record centering; keep the bootstrap uncertainty conditional.

The sources address different parts of the same inference problem. A continuous-time response model defines the quantities that can be identified; an information objective then assigns precision to local directions or predictions. A profile explores beyond that local quadratic approximation. The original TC-Lab model's exact equivalent family remains indistinguishable under every input in the stated experiment, so selecting a larger determinant cannot resolve its physical ambiguity.

Covariance estimation and resampling also answer different questions. HAC uses residual score products to estimate estimator variability; a circular bootstrap generates new records from resampled residual blocks. Both require assumptions about the errors around the fitted mean. A systematic error on a different experiment is evidence about mean transfer and experimental conditions, rather than evidence that either covariance procedure has measured that discrepancy.

The next measurement questions follow from these limits. An independent heater-power calibration would address the absolute conductance–capacity scale. Ambient and command prehistory records would constrain startup nuisance assumptions. A measurement equation connecting the second sensor to a modeled state is needed before using it to select an original physical family member. The adaptive-design journal paper and the unavailable bootstrap source listed in the search record remain reading leads, without theorem claims assigned to unread versions.

F Literature coverage and claim audit

The review used the configured lab literature skill and project-oriented searches of the local library, followed by primary-source web discovery. It is a focused review, without an exhaustive database search or a formal systematic-review protocol. Search passages that only list another work in their references were excluded as support for that work's claims.

The source record `literature/source_review.json` retains paper identities, source events where available, physical-page locators, and claim limitations. The record `literature/acquire`

`d_sources.json` identifies downloaded PDFs by their hashes. Full paper text and PDFs remain outside this repository.

For Raue et al. (2009), the library paper identifier is `6925c25f...ae8160`; the inspected profile construction is on physical PDF page 4, printed page 1926, Eq. (10). We checked that equation against the rendered page. The journal source by Hjalmarsson (2009) was checked at physical pages 8–9, printed pages 282–283, for confidence regions, identifiable combinations, and the assumptions behind prediction-error information. Those two pages were also inspected as rendered images because the extraction engines disagree on equation symbols. The journal source by Hurtado and Kirosingh (2019) was checked at physical pages 11–12, printed pages 1841–1842, for the chain equations and their integral-equation equivalence, with rendered page inspection of the mathematical passages.

The journal PDF of Newey and West (1987) was an image-only scan. We read physical pages 1–3, printed pages 703–705, with optical character recognition and checked the rendered pages. Equation (5) on physical page 2 gives the Bartlett weights; Theorem 2 on page 3 states assumptions for consistency. The scanned equation symbols were checked visually rather than copied from the recognition output.

The chapter by Garnier et al. (2008) was inspected at physical pages 27–29 of the stored book, printed pages 1–3. Its byline includes Peter C. Young in addition to the two book editors. The residual decomposition is on physical page 28, printed page 2. Chapter and editor roles are retained separately in the reference. Physical-to-printed page offsets were checked at each passage rather than extrapolated to the whole supplied PDF. Contextual chapter readings remain separate leads in the source record.

The journal scan of Politis and Romano (1994) was read with optical character recognition and all three cited pages were checked as rendered images. Physical pages 1–3 are printed pages 1303–1305. The circle convention is specified on physical page 1; the fixed-length relative is discussed in Remark 2 on physical page 3. The geometric-block stationarity and consistency results are not ascribed to our fixed-length nonlinear residual bootstrap.

For Manchester (2010), we inspected the eight-page author preprint. We cite its identifier, `arXiv:1009.5614`, and retain the distinct six-page conference version as an unread publication record. Its input-information formulation and optimization assumptions were inspected in the body. The downloaded adaptive-input-design preprint changes title and author order relative to its published journal record. Access to the journal body remains unresolved, and it is not cited for a theorem. These version and access limitations remain in the source records.

The exact expanded manuscript, citation inventory and claim assessments are bound in the private evidence packet. Scoped public receipts record the draft hash, inspected source identities, available fine-evidence bindings and explicit gaps for externally inspected PDFs that have no ingested carrier. Two cited works have ingested carriers; five primary PDFs were inspected externally. Source availability, quote fidelity, claim support and human scientific review remain separate statuses. Mechanical checks and agent assessments do not constitute human acceptance of the manuscript.